Design of Embedded Systems with RaspberryPI

Version 1.0

Design of Embedded Systems with RaspberryPI.

Build: 07.05.2026
Author: Mariano Ruiz

Table of Contents

Contents:

Building Embedded Linux for Raspberry PI

1. Acronyms

CPU
Central Processing Unit

EABI
Extended Application Binary Interface

EHCI
Enhanced Host Controller Interface

GPU
Graphical Processing Unit

I/O
Input and Output

MMC
Multimedia card

NAND
Flash memory type for fast sequential read and write

OS
Operating system

PCI
Peripheral Component Interconnect – computer bus standard

PCI Express
Peripheral Component Interconnect Express

UART
Universal Asynchronous Receiver Transmitter

USB
Universal Serial Bus

2. Getting started with RPI

This document describes the essential steps students have to follow to boot the Raspberry Pi with a Linux-based distribution such as Raspberry PI OS. After performing the operating system installation, the student must investigate the software tools installed and their potential. Finally, the student must implement a software application that measures 3D acceleration using a transducer. The following variants of application are proposed to the student:

1. Develop a simple console program developed in the C language.

2. (Optional) Develop an equivalent application using Python.

Note

[Time to complete the laboratory]: The time necessary to complete steps 1 to 3 in this tutorial is approximately 4 hours.

2.1. Reference Documentation

The following documentation has to be read and checked.

1. http://www.raspberrypi.org/help/. This web contains some guides and documents covering a lot of topics

2. https://projects.raspberrypi.org/en . RPI application examples

3. http://www.python.org. Phyton resources, tutorial, and reference

4. https://www.raspberrypi.org/courses/learn-python

5. http://i2c.info/i2c-bus-specification

2.2. Operating System Installation

2.2.1. Downloading and installing Raspberry PI OS

Several Linux and Android distributions can be installed on the Raspberry Pi platform. These Operating System distributions also include multiple software applications and tools such as text editors, music players, video editing and playing, games, development tools, etc.

Follow the steps below to install the Raspbian Operating System:

1. Download the Raspberry PI Imager tool for your specific OS.
2. Execute the application, and a window will be displayed.
    1. First, choose the Raspberry PI hardware, in this case, **Raspberry PI 4 Model B** (see Fig. 2.1).
    2. Choose the Operating System (see Fig. 2.2) and **Raspberry PI OS 64 bits version**.
    3. Next, choose the storage device where the OS image will be written (see Fig. 2.3).
    4. Now, set the device hostname (see Fig. 2.4).
    5. Then, set the localisation options (see Fig. 2.5). Set the keyboard to your language.
    6. Next, set the user configuration (see Fig. 2.6). Set the username to **rpi-student** and the password to **rpi**.
    7. Now, set the WiFi configuration (see Fig. 2.7). Set the SSID to **RPI** and the password to **UPMRPI2024**.
    8. Enable SSH so you can connect remotely to the RPI (see Fig. 2.8).
    9. Finally, write the image to the SD card (see Fig. 2.9).

⚠ Warning

[VERY IMPORTANT]: Confirm the destination device where the image is burned because this information is not recoverable.

1. Wait until the program ends writing the SD Card **(this can take up to one hour depending on your SD card category and model)**.

Note

You can find more information about the Raspberry Pi Imager tool in the official documentation

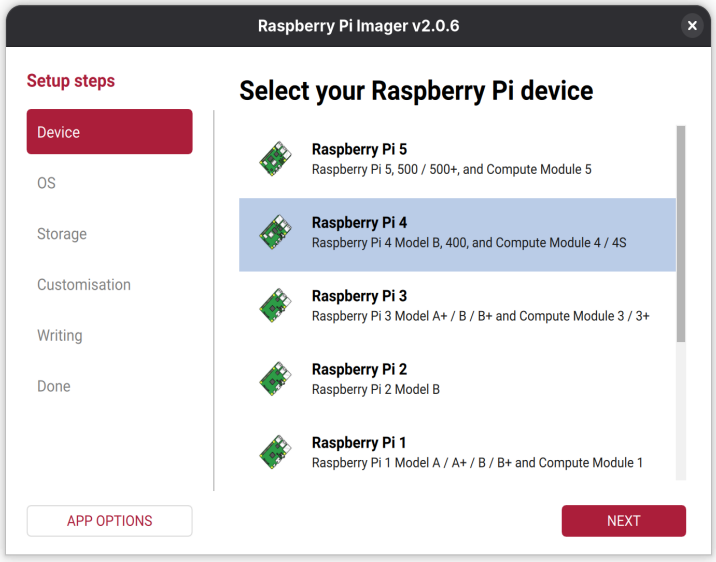


Fig. 2.1 Raspberry Pi Imager HW model

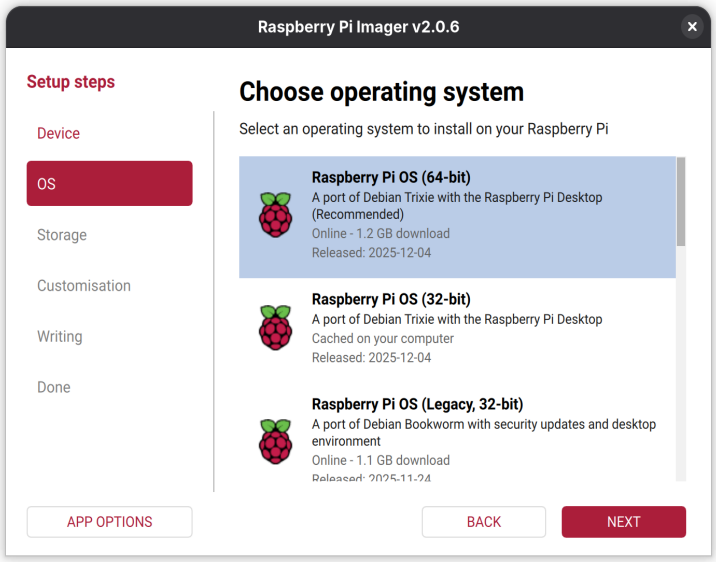


Fig. 2.2 Raspberry Pi Imager OS selection

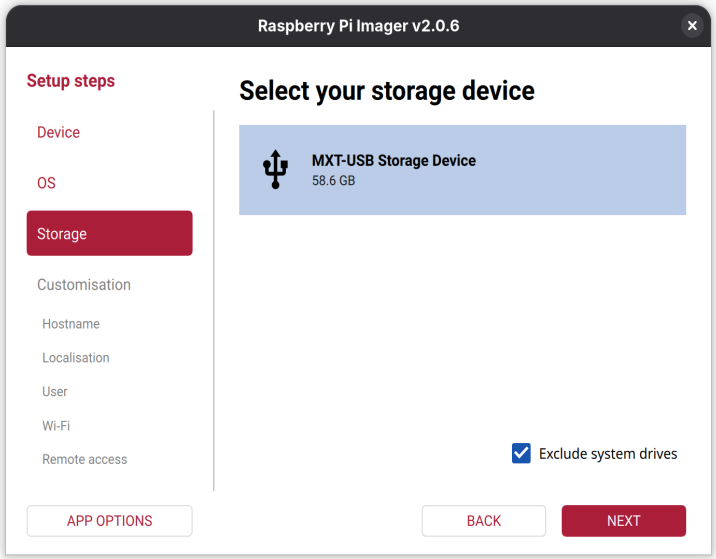


Fig. 2.3 Raspberry Pi Imager Storage selection

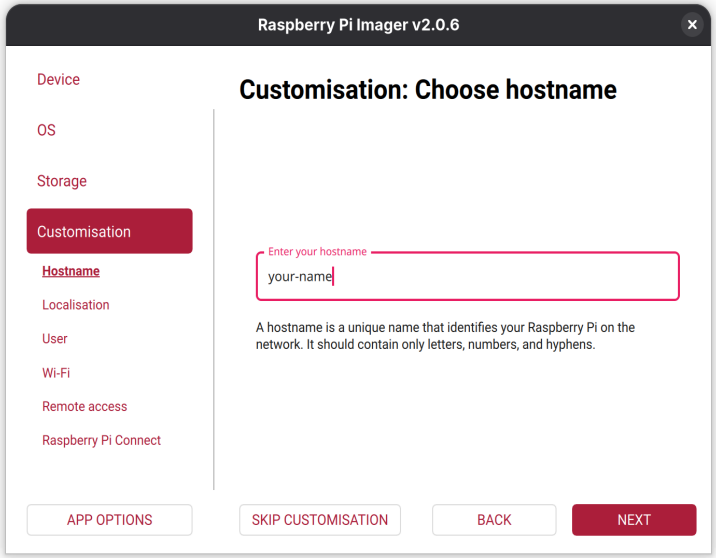


Fig. 2.4 Raspberry Pi Imager device hostname

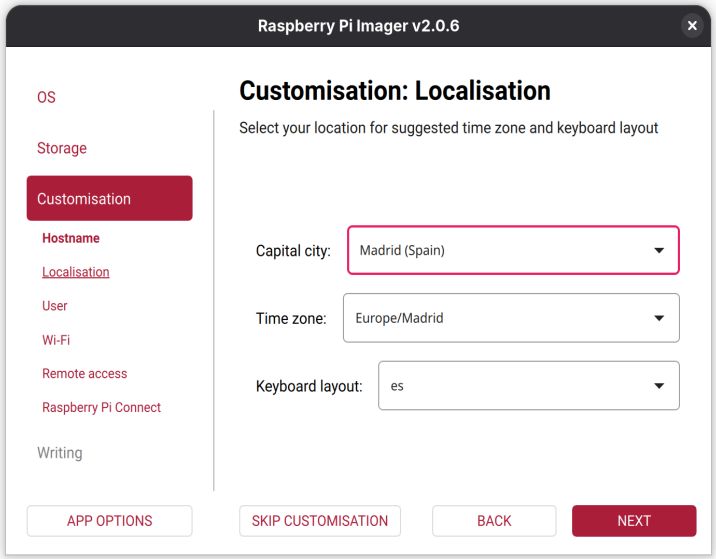


Fig. 2.5 Raspberry Pi Imager localisation and keyboard settings

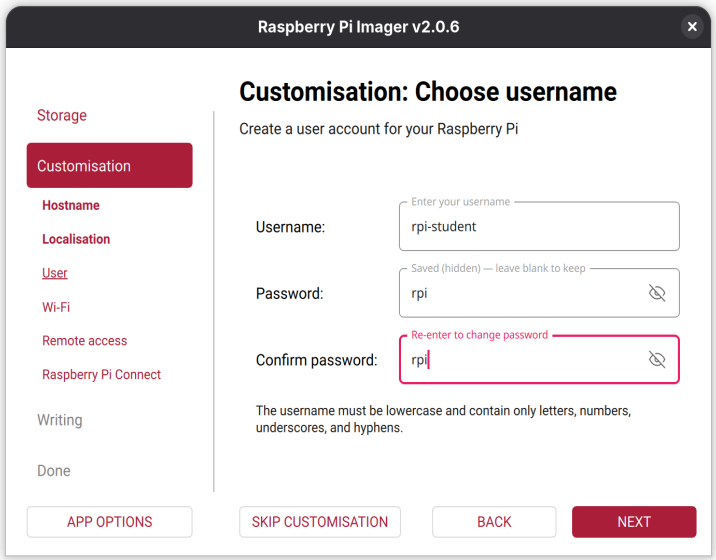


Fig. 2.6 Raspberry Pi Imager user settings

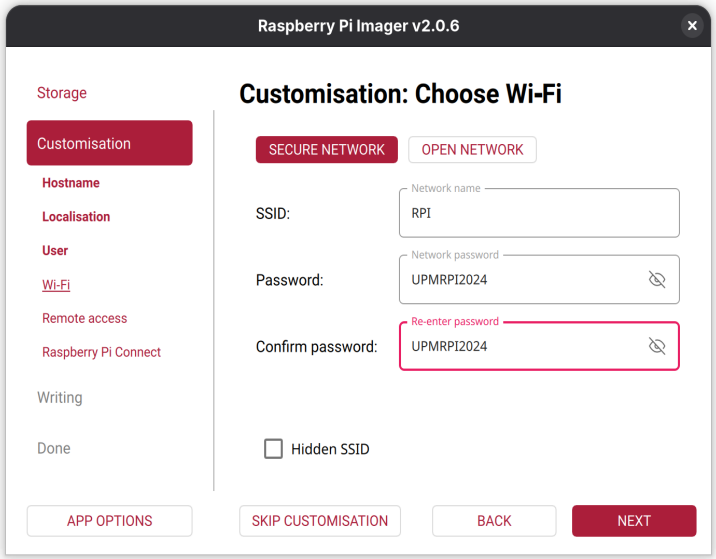


Fig. 2.7 Raspberry Pi Imager WiFi settings

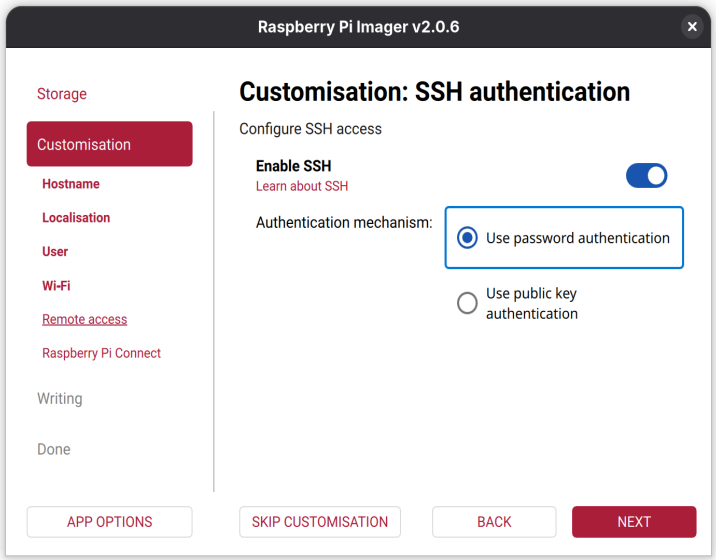


Fig. 2.8 Raspberry Pi Imager SSH settings

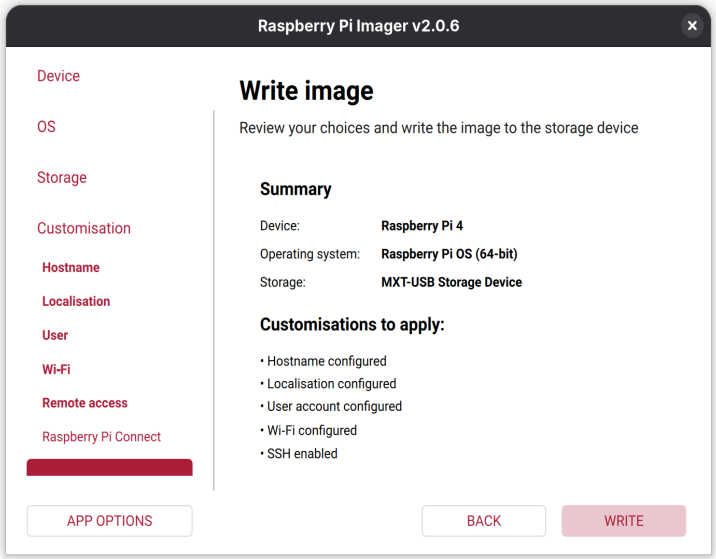


Fig. 2.9 Raspberry Pi Imager write SD card

2.2.2. Setting up the Raspberry Pi

We need to interact with the Raspberry Pi for the activities described in this document. There are three ways to establish this interaction:

- The easiest one is connecting an HDMI monitor, a USB keyboard, and a USB mouse. This configuration allows us to use the Raspberry Pi as we would with a regular computer. Once the RPI is booted, we can set the WiFi connections and other settings such as enable the use of a serial line.

- Opening an SSH session between your computer and the Raspberry Pi without connecting additional peripherals. This kind of configuration is known as headless start (use the *rpi-student* login).

  The Raspberry Pi must be connected to the same network as your computer. First, you need to discover the Raspberry Pi IP address (see the section Discovering the RPI IP address).

- Use the serial USB-TTL cable that you need to connect to the GPIO expansion connector. This requires first to enable the serial line in the Raspberry PI OS using the raspi-config tool. The connection of the USB-TTL cable is depicted in Fig. 2.10.

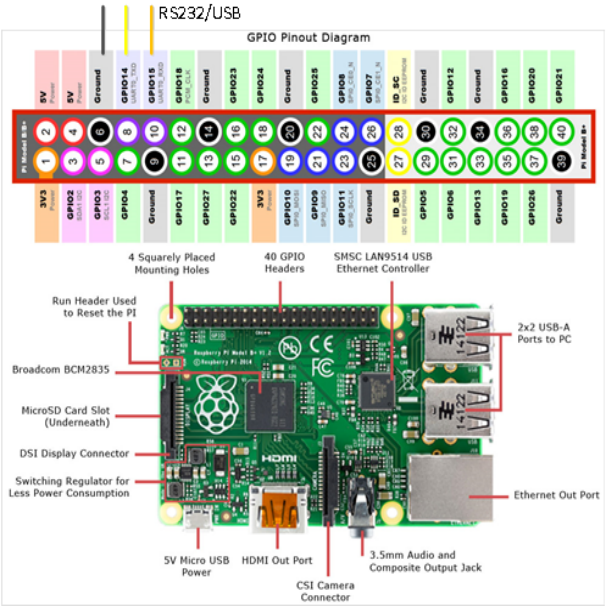


Fig. 2.10 Detail of expansion connector highlighting the USB-TTL RS232

2.2.3. Reviewing the SD card content

1. Plug in the SD card into the computer.

2. You should now see one new drive in the System Explorer (Windows)/Nautilus (Ubuntu) named “boot”. In Linux you will see two partitions mounted in your system. Open the partition/unit identified as “boot”.

Note

[VERY IMPORTANT]: This is the partition used by Raspberry to make the necessary hardware configuration before starting the operating system. Beware of not deleting or modifying files other than those described in this manual, or the Raspberry may not boot.

1. List the content of the boot partition (FAT32) and identify the files as in Fig. 2.11 (Ubuntu Linux).

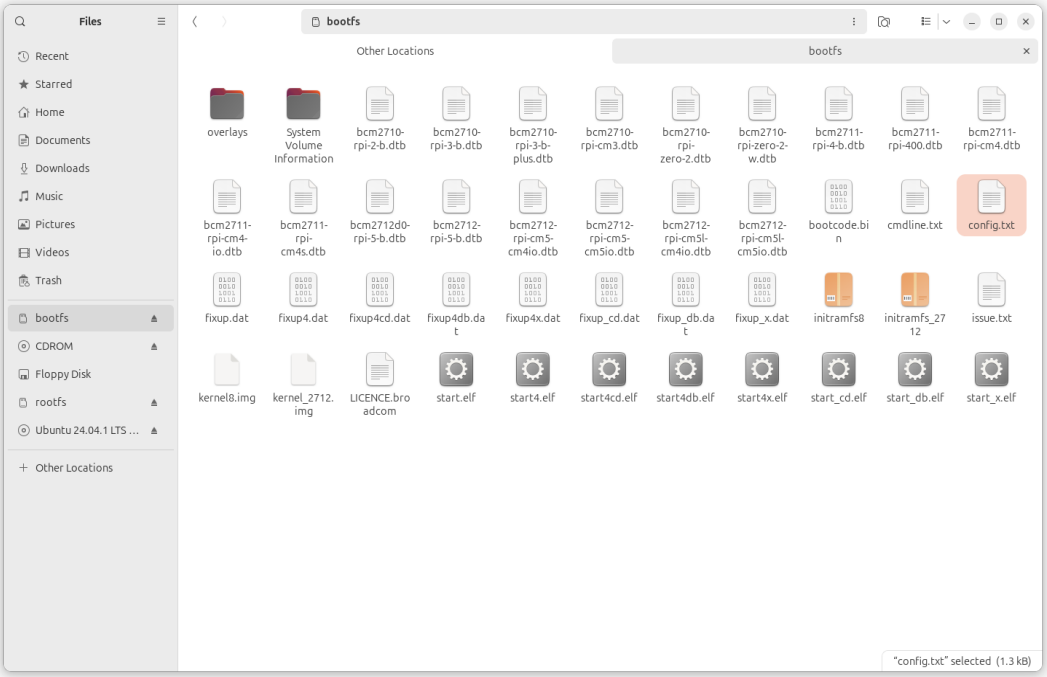


Fig. 2.11 Files available in the FAT32 partition

2.2.4. Booting the RPI

Insert the SD card and connect the power supply and all other cables needed. The first time you boot your RPI you will see how the RPI is configured (this information is only displayed in the HDMI monitor) and rebooted. Wait 1 or 2 minutes (this depends again on your SD card category) until the RPI is completely booted.

Note

[Security]: Having the default user and password supposes a security issue, more if the SSH sessions are enabled. You can always change the password by executing the command *passwd* once logged in.

2.2.5. Discovering the RPI IP address

To open an SSH session from your computer to the Raspberry Pi you need to know its IP address. There are multiple ways to discover the Raspberry Pi IP address:

1. Use an IP Scanner utility, and scan in the range of your network. For example, you can quickly identify a Raspberry Pi as they use the hostname “raspberrypi.local” by default. This tool cannot be used at the UPM lab.

2. If you know your Raspberry Pi MAC address, you can set the DHCP server to assign a static IP to the Raspberry Pi MAC. This is typically done in the router configuration. Please, consider that Ethernet and WiFi ports have different MAC addresses. In the laboratory, your Raspberry can have only a dynamically assigned IP in the WIFI network with SSID RPI (ask your instructor to verify the configuration).

3. You could open an RS232 serial connection in the Raspberry Pi GPIO ports and use the command line (*ifconfig* command) to obtain the IP or set the desired IP and netmask.

2.2.6. Raspberry Pi OS Update

You can update and upgrade the OS if you have an internet connection. Please, do not update the OS during the classes, as this operation will require some time. You can update the OS by running the following commands:

Listing 2.1 RPI OS update

```
$ sudo apt update
$ sudo apt upgrade
```

2.3. Integration of a 3-axis accelerometer with Raspberry-Pi

2.3.1. Specifications

The student must investigate functions and commands and implement a C program to get the following requirements:

1. Show the values of the 3-axis acceleration obtained from the MPU-6000 I2C sensor.

2. The measurement readings are shown every 3 seconds.

2.3.2. Suggested Improvements

- Analyses the problem of offsets in the values returned. Define and implement a method to get corrected values
- Get stop the program if “CRTL-C” is typed.

2.3.3. Optional: Python Script

For advanced students, the implementation of this exercise in Python is proposed. For this, you can use any of the multiple Python modules that gain access to I2C interface in the Raspberry Pi.

2.4. Compiling & linking C program in Linux

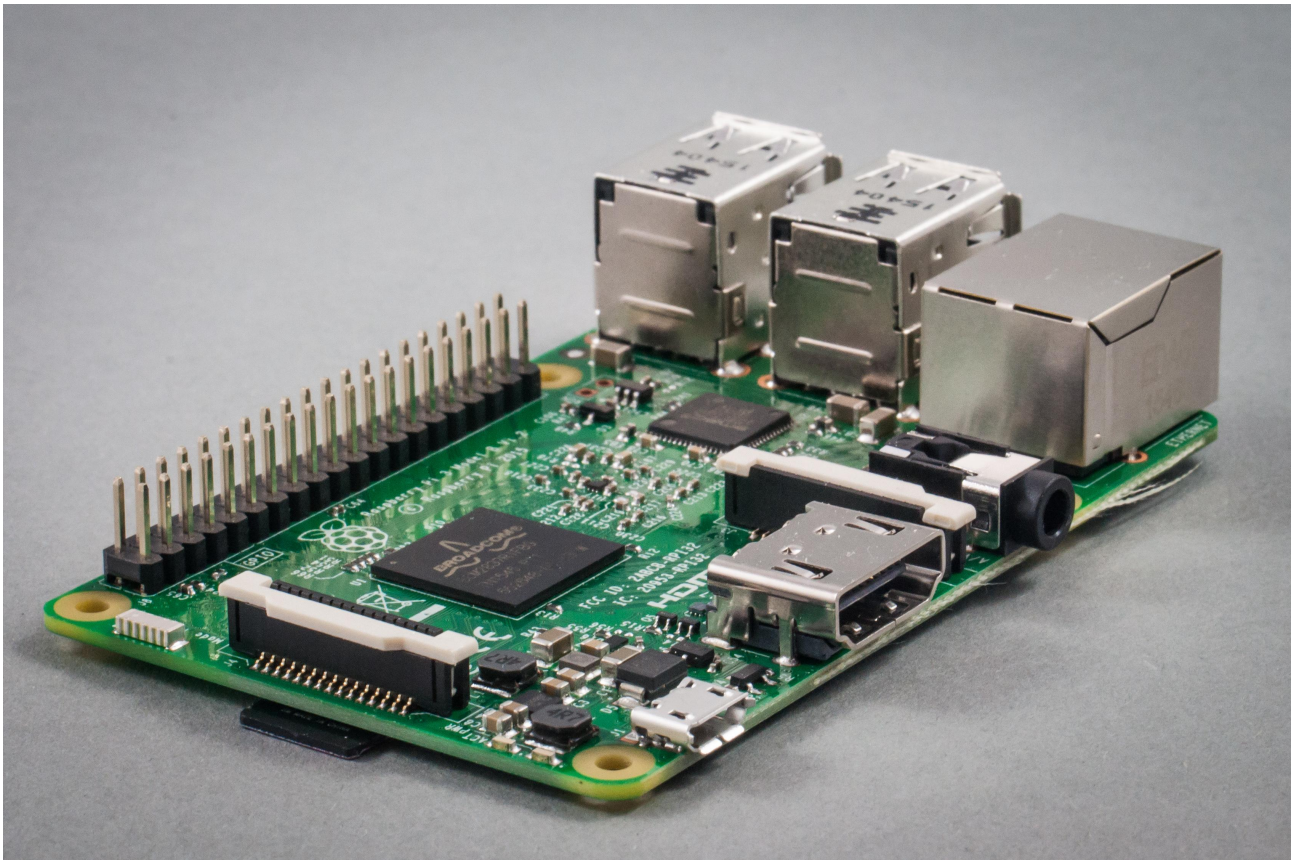
In Linux, C/c++ programs are compiled and linked using the “make” command. This command searches a “Makefile”. A “Makefile” is a text file with the necessary information to compile and link and must be in the same directory of *myprogram.c*. The result will be *myprogram.o* and *myprogram* (executable).

Listing 2.2 Detail Makefile

```
DEBUG= -O2 #Debugging Level

CC= gcc #Compiler command
INCLUDE= -I/usr/local/include #Include directory
CFLAGS= $(DEBUG) -Wall $(INCLUDE) -Winline #Compiler Flags
LDFLAGS= -L/usr/local/lib #Linker Flags

LIBS= -lpthread -lm #Libraries used if needed

SRC = myprogram.c

OBJ  = $(SRC:.c=.o)
BIN  = $(SRC:.c=)

$(BIN):       $(OBJ)
     @echo [link] $@
     $(CC) -o $@ $< $(LDFLAGS) $(LIBS)
.c.o:
     @echo [Compile] $<
     $(CC) -c $(CFLAGS) $< -o $@

clean:
     @rm -f $(OBJ) $(BIN)
```

2.5. Questions to be reported to instructors

1. Explain the content of the *config.txt* file in the FAT32 partition of the uSDCARD of your RPI
2. How the I2C interface is enabled? Is there any change in the config.txt file?
3. Could you describe the commands used to compile a C/C++ program in Linux?
4. What is a library? What is the difference between a static and a shared library?
5. Summarize the utility of the *makefile* file and the make command.
6. What is the preferred utility to debug a C program in Linux?
7. How can you obtain the serial number of your Raspberry Pi? Is there any relation between the serial number and the MAC address?



3. Embedded Linux With RPI

3.1. Document Overview

This document describes the steps to develop an embedded Linux-based system using the Raspberry PI board. The document has been specifically written to use a Raspberry PI development system based on the BCM2837 processor. All the software elements used have a GPL license.

Note

The time necessary to complete all the tutorial steps is approximately 8 hours.

Read all the instructions carefully before executing the practical part; Otherwise, you will find errors, which are probably unpredicted. In parallel, you need to review the slides available at the Moodle site or at [RD1]

3.2. References

1. Embedded Linux system development. Slides
2. https://bootlin.com/training/embedded-linux/
3. Mastering Embedded Linux Programming - Second Edition. Packt. https://www.packtpub.com/product/mastering-embedded-linux-programming-second-edition/9781787283282
4. Raspberry-Pi User Guide. Reference Manual.

   www.myraspberry-pi.org/wp-content/.../Raspberry.Pi_.User_.Guide_.pdf

3.3. Building Linux using buildroot

3.3.1. Elements needed for the execution of these LABS

In order to execute this lab properly, you need the following elements:

1. The VMware player software version 17.6.1 or above. Available at Broadcom website (free download and use but you need to register). This software has already been installed on the laboratory desktop computer.

2. A VMWare virtual machine with Ubuntu 24.04 and all the software packages installed is already available on the Desktop. This virtual machine is available for your personal use. If you want to set up your virtual machine by yourself, follow the instructions provided in Annex I.

3. A Raspberry Pi, accessories and a USB cable are available at the laboratory.

4. Basic knowledge of Linux commands.

3.4. Starting the VMware

Start VMware Player and open the RPI Virtual Machine. Wait until the welcome screen is displayed (see Fig. 3.1 and Fig. 3.2). Login as "*ubuntu*" user using the password "ubuntu".

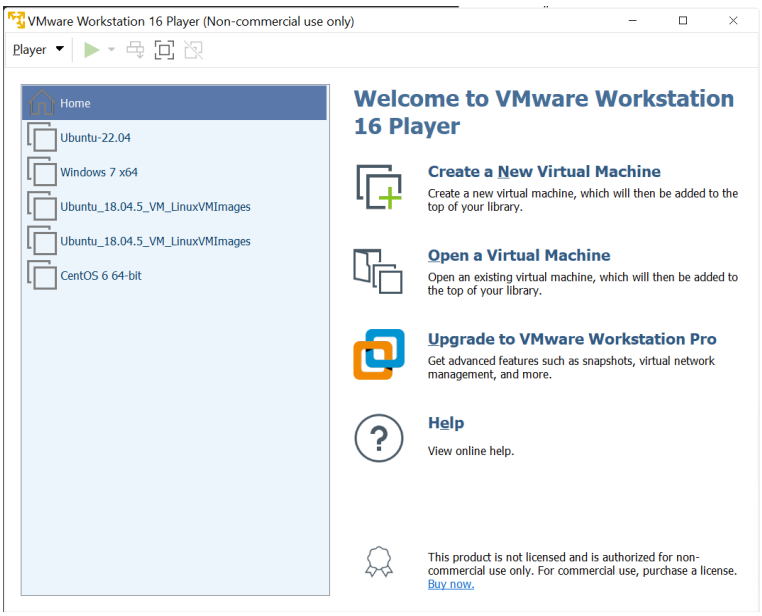


Fig. 3.1 Main screen of VMware player with some VM available to be executed

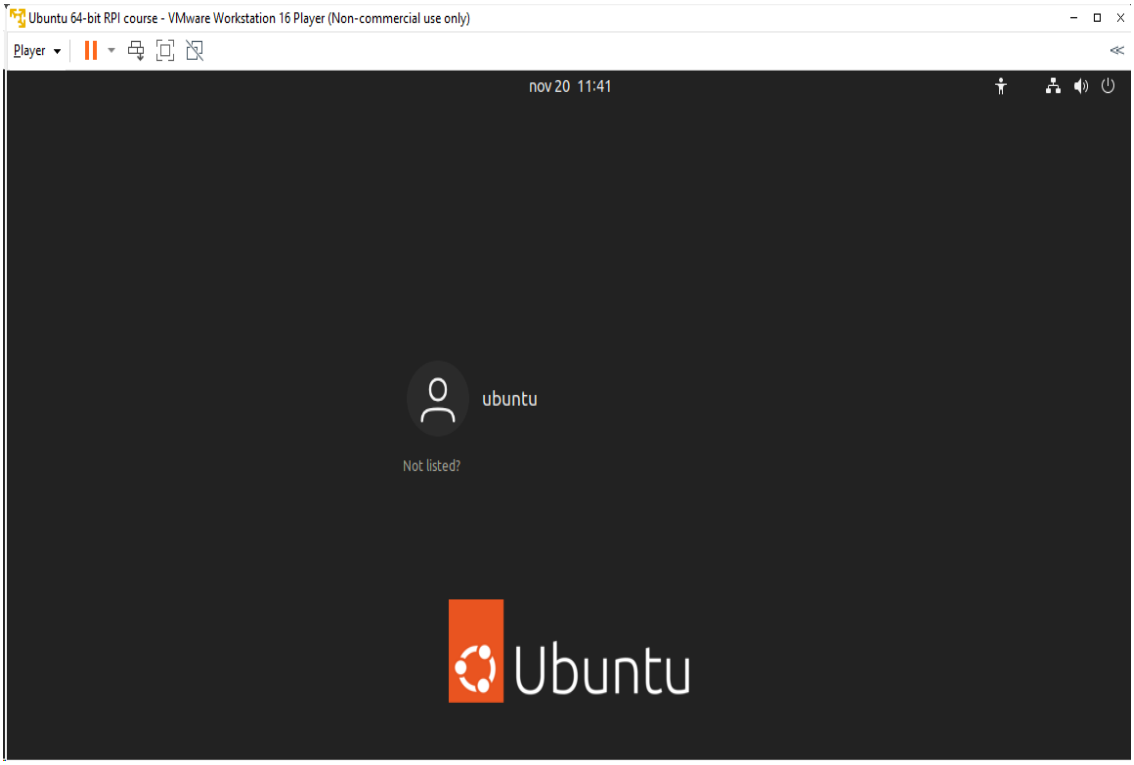


Fig. 3.2 Ubuntu Virtual Machine login screen.

Open the **Firefox** web browser and download from https://buildroot.org/, the version identified as **buildroot2025.02.9** (use the download link, see Fig. 3.3, and navigate searching for earlier releases if necessary, https://buildroot.org/downloads/). Save the file to the **Documents** folder in your account (Fig. 3.4).

Fig. 3.3 Buildroot home page.

Buildroot is a tool to generate embedded Linux systems in our PC, and then this Linux will be installed in the target.

Index of /downloads/

Name	Last Modified↑	Size:	Type:
buildroot-2024.02.7.tar.xz.sign	2024-Oct-21 09:14:07	0.6K	application/octet-stream
buildroot-2024.02.7.tar.xz	2024-Oct-21 09:14:06	5.2M	application/x-xz
buildroot-2024.02.7.tar.gz.sign	2024-Oct-21 09:14:04	0.6K	application/octet-stream
buildroot-2024.02.7.tar.gz	2024-Oct-21 09:14:04	7.0M	application/x-gtar-compressed
manual/	2024-Oct-20 16:25:45	-	Directory
buildroot-2024.08.1.tar.xz.sign	2024-Oct-20 16:25:18	0.6K	application/octet-stream
buildroot-2024.08.1.tar.xz	2024-Oct-20 16:25:18	5.3M	application/x-xz
buildroot-2024.08.1.tar.gz.sign	2024-Oct-20 16:25:15	0.6K	application/octet-stream
buildroot-2024.08.1.tar.gz	2024-Oct-20 16:25:15	7.2M	application/x-gtar-compressed
buildroot-2024.02.6.tar.xz.sign	2024-Sep-09 17:07:01	0.6K	application/octet-stream
buildroot-2024.02.6.tar.xz	2024-Sep-09 17:07:01	5.2M	application/x-xz
buildroot-2024.02.6.tar.gz.sign	2024-Sep-09 17:06:59	0.6K	application/octet-stream
buildroot-2024.02.6.tar.gz	2024-Sep-09 17:06:58	7.0M	application/x-gtar-compressed
buildroot-2024.05.3.tar.xz.sign	2024-Sep-09 09:45:14	0.6K	application/octet-stream
buildroot-2024.05.3.tar.xz	2024-Sep-09 09:45:13	5.2M	application/x-xz
buildroot-2024.05.3.tar.gz.sign	2024-Sep-09 09:45:12	0.6K	application/octet-stream
buildroot-2024.05.3.tar.gz	2024-Sep-09 09:45:11	7.1M	application/x-gtar-compressed
buildroot-2024.08.tar.xz.sign	2024-Sep-06 15:38:59	0.5K	application/octet-stream
buildroot-2024.08.tar.xz	2024-Sep-06 15:38:58	5.3M	application/x-xz
buildroot-2024.08.tar.gz.sign	2024-Sep-06 15:38:47	0.5K	application/octet-stream
buildroot-2024.08.tar.gz	2024-Sep-06 15:38:46	7.2M	application/x-gtar-compressed
buildroot-2024.08-rc3.tar.xz.sign	2024-Sep-01 21:51:43	0.6K	application/octet-stream
buildroot-2024.08-rc3.tar.xz	2024-Sep-01 21:51:42	5.3M	application/x-xz
buildroot-2024.08-rc3.tar.gz.sign	2024-Sep-01 21:51:32	0.6K	application/octet-stream
buildroot-2024.08-rc3.tar.gz	2024-Sep-01 21:51:30	7.2M	application/x-gtar-compressed
buildroot-2024.08-rc2.tar.xz.sign	2024-Aug-22 18:24:36	0.6K	application/octet-stream
buildroot-2024.08-rc2.tar.xz	2024-Aug-22 18:24:35	5.3M	application/x-xz
buildroot-2024.08-rc2.tar.gz.sign	2024-Aug-22 18:24:33	0.6K	application/octet-stream
buildroot-2024.08-rc2.tar.gz	2024-Aug-22 18:24:33	7.2M	application/x-gtar-compressed
buildroot-2024.02.5.tar.xz.sign	2024-Aug-14 13:35:48	0.6K	application/octet-stream
buildroot-2024.02.5.tar.xz	2024-Aug-14 13:35:47	5.2M	application/x-xz
buildroot-2024.02.5.tar.gz.sign	2024-Aug-14 13:35:46	0.6K	application/octet-stream
buildroot-2024.02.5.tar.gz	2024-Aug-14 13:35:45	7.0M	application/x-gtar-compressed
Vagrantfile	2024-Aug-14 11:08:14	1.7K	application/octet-stream
buildroot-2024.05.2.tar.xz.sign	2024-Aug-14 11:07:57	0.6K	application/octet-stream

Fig. 3.4 Example of Downloading Buildroot source code.

Create a folder “rpi” in “Documents”. Copy the file to the “Documents/rpi” folder and decompress the file (Fig. 3.5).

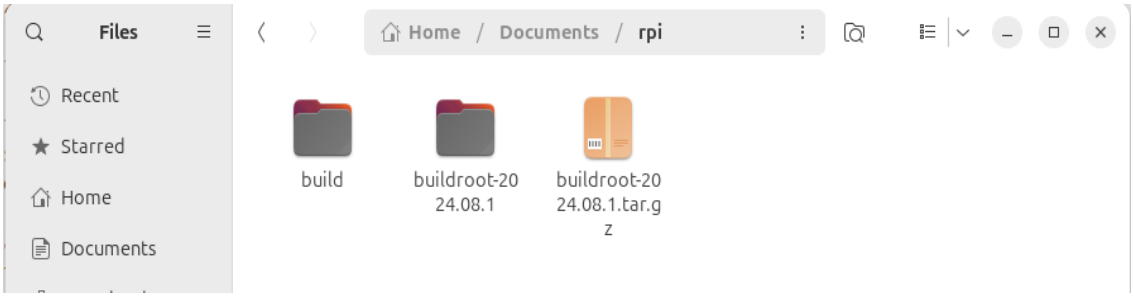


Fig. 3.5 Buildroot folder (the folder name depends on the version downloaded).

Right-click in the window and execute “Open in Terminal” or execute the Terminal application from Dash home as shown in Fig. 3.6 (if “Open in Terminal” is not available, search how to install it in Ubuntu).

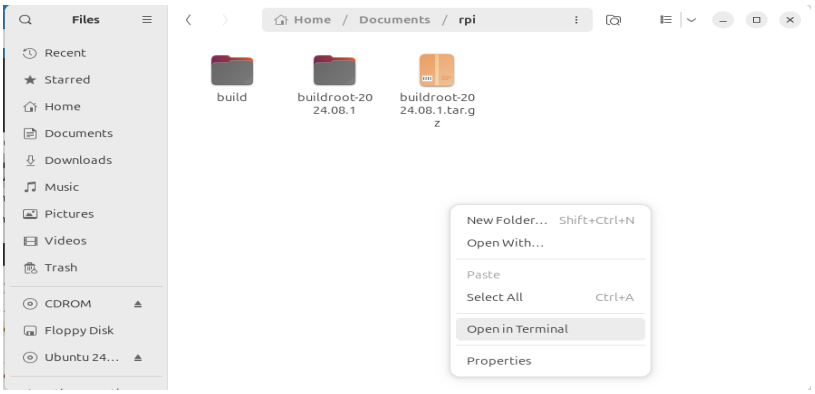


Fig. 3.6 Terminal application

In some seconds, a command window is displayed. Then, execute these commands:

```
$ mkdir build
$ cd build
```

```
$ make O=$PWD -C /home/ubuntu/Documents/rpi/buildroot-2025.02.9/
menuconfig
```

Important

The **build** folder will contain all the compilation of the embedded Linux generated. Remember to change to this folder before doing any operation with buildroot

Important

For this course, you will need to become familiar with the Linux Terminal use. On the Moodle site of this course, you can find a cheat sheet with the basic Linux commands.

Tip

In a Linux terminal, the “TAB” key helps you to autocomplete the commands, folders, and file names.

In some seconds, you will see a new window similar to Fig. 3.7.

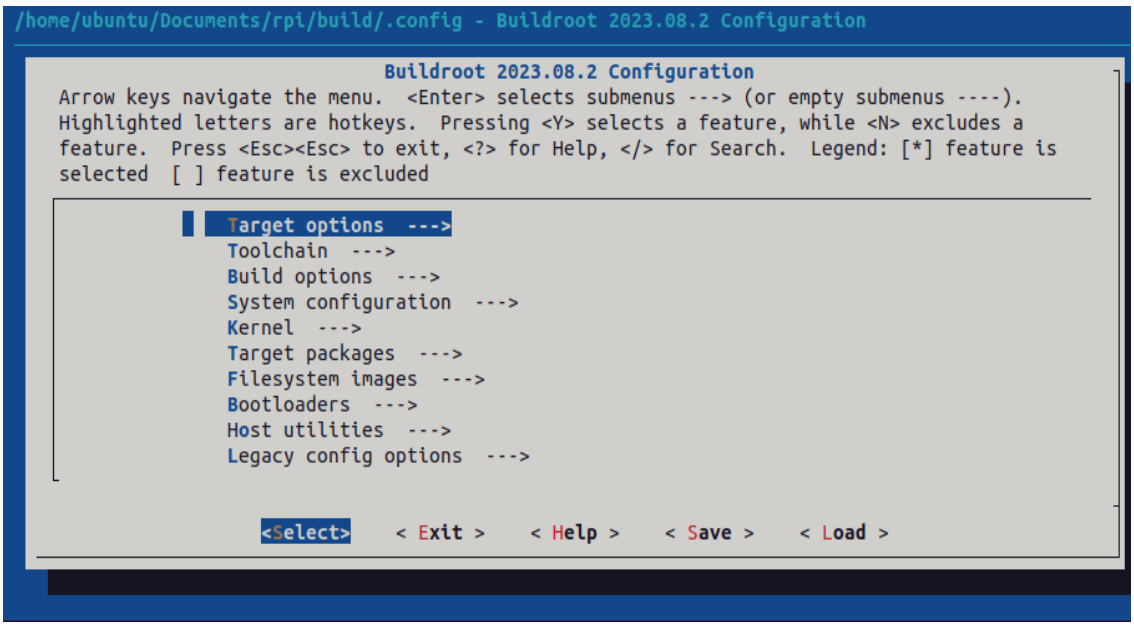


Fig. 3.7 Buildroot setup screen.

3.5. Configuring Buildroot for RPI4

Once the **Buildroot** configuration is started, it is necessary to configure the different items. You need to navigate the different menus and select the installation elements. Table I contains the specific configuration of **Buildroot** for installing it in the Raspberry Pi. Depending on the downloaded version, the organization and the items displayed can differ. If an item of buildroot configuration does not appear in the Table I leaves it with its default value.

Important

The Buildroot configuration is an iterative process. In order to set up your embedded Linux system, you will need to execute the configuration several times.

Warning

The tables have three columns. Check that you understand all the content shown.

3.5.1. Target Options

This is the selection of the processor to use (Table 3.1).

Table 3.1 Target Options

Target Architecture	AArch64 (little endian)	ARM 64 bits
Target Architecture Variants.	Cortex-A72	
Floating Point Strategy	VFPv4	
MMU Page Size	4KB	
Target Binary Format	ELF	

3.5.2. Toolchain

Cross Compiler, linker, and libraries to be built to compile our embedded application. Select the options shown in the following table (Table 3.2).

Table 3.2 Toolchain

Toolchain type	Buildroot toolchain	The Embedded Linux System will be compiled with tools integrated into Buildroot
Custom toolchain vendor name.	buildroot	
C library	glibc	Library containing the typical C libraries used in Linux environments (stdlib, stdio, etc)
Kernel Headers	Same as kernel being built	Include files for kernel
Kernel Header Options	6.6.x kernel headers	
Binutils Version	2.43.1	Binutils contains tools to manage the binary files obtained in the compilation of the different applications
GCC compiler Version	gcc 13.x	GCC tools version to be installed
Enable C++ support	Yes.	Including support for C++ programming, compiling, and linking.
Build cross gdb for the host	Yes.	Includes the support for GDB.

Add Python support	Yes
GDB debugger version	gdb 15.x

3.5.3. Build options

How Buildroot will build the code. Leave the default values.

3.5.4. System Configuration

Here you can define the basic configuration of the embedded Linux to generate and specific scripts to add additional functionality (Table 3.3).

Table 3.3 System-configuration

Root FS skeleton	Default target skeleton.	Linux folder filesystem organization for skeleton the embedded system
System hostname	**buildroot**	Name of the embedded system
System Banner	**Linux RPI 4**	Banner.
Passwords encoding	sha 256	
Init System	Busybox	
/dev management	Dynamic using devtmpfs + mdev	
Path to permissions for table	**system/device_table.txt**	
	yes	

Enable root login with password		
Root password	rpi	
Busybox' default shell	/bin/sh	
Run a getty after boot	tty PORT: **console**. Baudrate: keep kernel default. TERM environment variable: vt100	
remount root filesystem read write during boot	Yes	
Network interface to configure through DHCP	eth0	
Set the system's default PATH	/bin:/sbin:/usr/bin:/usr/sbin	
Purge unwanted locales	yes	
Leave the default values for all others		
Custom scripts to run path **before** creating filesystem images	**your path**/buildroot-2025.02.9/board/raspberrypi4-64/post-build.sh	your path -> "/home/...." os where you have decompressed buildroot
Custom scripts to run inside the fakeroot environment		

Custom scripts to run **after** creating filesystem images	**your path**/buildroot-2025.02.9/board/raspberrypi4-64/post-image.sh

3.5.5. Linux Kernel

This is the configuration of the Linux kernel. The specific location and version is specified among other parameters (Table 3.4). See third column for details of git repo to use.

Table 3.4 kernel-configuration

Kernel Version	Custom tarball.	$(call github,raspberrypi,linux,576cc10e1ed50a9eacffc7a05c796051d7343ea4)/linux-576cc10e1ed50a9eacffc7a05c796051d7343ea4.tar.gz
Kernel configuration	Using and intree defconfig file	
Defconfigname	bcm2711	This file contains the specific configuration of the kernel for the RPI
Kernel binary format	Image	
Kernel compression format	gzip compression	
Build a Device Tree Blob (DTB)	Yes	
Intree Device Tree Source file name	broadcom/bcm2711-rpi-4-b broadcom/bcm2711-rpi-400	

	broadcom/bcm2711-rpi-cm4 broadcom/bcm2711-rpi-cm4s
Need host OpenSSL	Yes
Linux kernel Extensions	Nothing
Linux Kernel Tools	Nothing

3.5.6. Target Packages

Target packages option allows to select the software elements that will be installed in the filesystem of the embedded Linux. Additionally, this option installs the busybox package that contains the basic Linux commands (Table 3.5). Buildroot creates the filesystem hierarchy following the Linux standard organization.

Table 3.5 Busybox and target packages

Busybox	yes
Busybox configuration file to use	package/busybox/busybox.config
Show packages that are also provided by busybox	Yes
Audio and video applications	Default values

Compresssors and decompressors	xz-utils	
Debugging, profiling and benchmark	**gdb, gdbserver, full debugger**	
Developments tools	Default values	
Filesystem and flash utilities	Default values	
Games	Default values	
Graphic libraries and applications (graphic/text)	Default values	
Hardware handling	**Firmware>rpifirmware rpi4 (default)**	Path to a file stores as boot/config.txt: **your path**/board/raspberrypi4-64/config_4_64bit.txt
Hardware handling	**Firmware>rpifirmware**	Path to a file stored as boot/cmdline.txt: **your path**/board/raspberrypi4-64/cmdline.txt
Hardware handling	**Firmware>rpifirmware**	**install DTB overlays**
Interpreters language and scripting Libraries	Python3	
Miscellaneous	Default Values	
Libraries	Default Values	

Networking applications	**ifupdown openssh**	**scripts,**
Package Managers	Default values	
Real Time, Shell and utilities	Default Values	
System Tools	kmod, kmod utilities	
Text Editor and Viewers	Default Values	

3.5.7. File System Images

This option selects the format of the root filesystem and the size (Table 3.6).

Table 3.6 Filesystem images

ext2/3/4 root filesystem	ext4	
filesystem label	rootfs	
exact size	**400M** Leave the other default values	Update this value with your specific needs
Compression method	No compression	

3.5.8. Boot-loaders

The Raspberry PI does not need an specific bootloader because it is incorporated in the firmware provided by Broadcom.

3.5.9. Host Utilities

Additional tools needed for ubuntu to create all the embedded images (Table 3.7).

Table 3.7 Host utilities

host environment setup	Yes
host genimage	Yes
host dosfstools	Yes
host kmod	Yes, support **xz-compressed modules**
host mtools	Yes

Once you have configured all the menus, you need to exit, saving the values (File->Quit).

3.6. Compiling buildroot

In the Terminal Window execute the following command (Listing 3.1):

Listing 3.1 Build Buildroot

```
$ make O=$PWD -C /home/ubuntu/Documents/rpi/buildroot-2025.02.9/
```

If everything is correct, you will see a final window similar to the one represented in Fig. 3.8.

⚠ Warning

In this step, buildroot will connect, using the internet, to different repositories. After downloading the code, Buildroot will compile the applications and generate a lot of files and folders. Depending on your internet speed access and the configuration chosen, this step could take up to **two hours**. If you have errors in the buildroot configuration, you could obtain errors in this compilation phase. Check your configuration correctly. Use “make O=$PWD -C /

home/ubuntu/Documents/rpi/buildroot-2025.02.9/ clean" to clean up your partial compilation.

Note

dl subfolder in your buildroot folder contains all the packages downloaded for the internet. If you want to move your buildroot configuration from one computer to another, avoiding the copy of the virtual machine, you can copy this folder. |

Warning

If your building process fails different reasons could be the origin. consider to use the following actions. Make a copy of your *.config* file (hidden file in Linux) to save your configuration.

Table 3.8 actions

make O=$PWD -C /home/ubuntu/Documents/rpi/ buildroot-2025.02.9/ clean	Build again buildroot
make O=$PWD -C /home/ubuntu/Documents/rpi/ buildroot-2025.02.9/ distclean	configure and build again buildroot

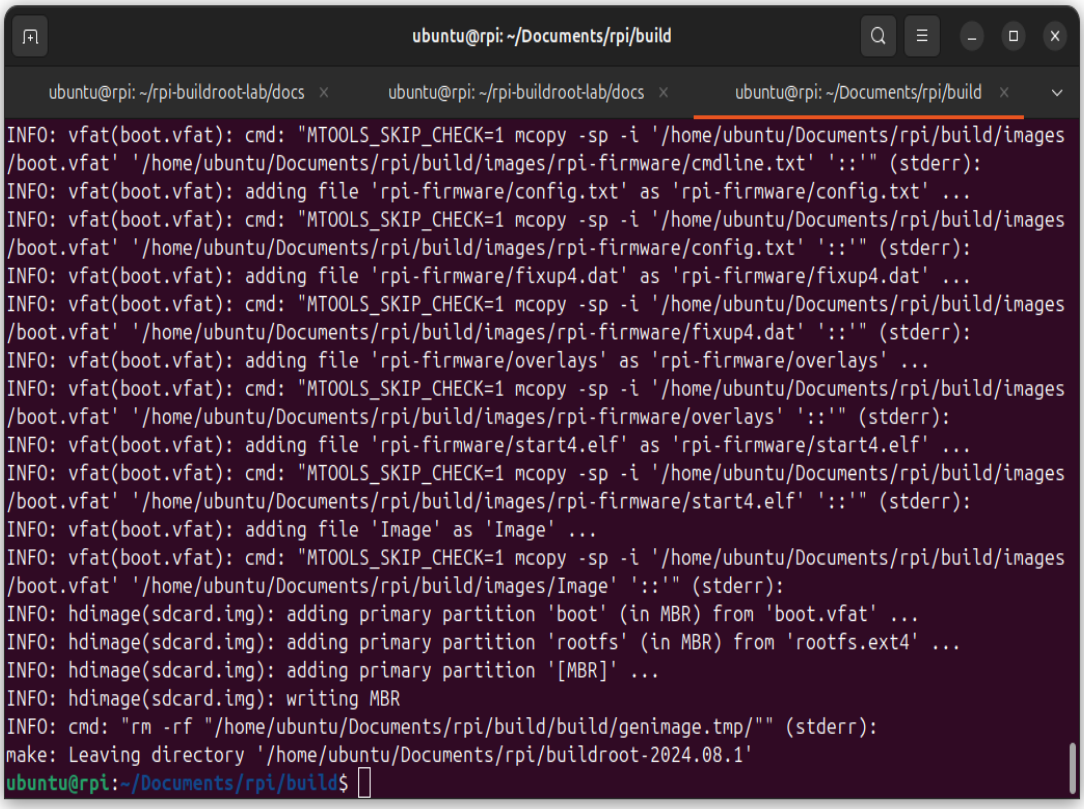


Fig. 3.8 Successful compilation and installation of Buildroot

Buildroot has generated some folders with different files and subfolders containing the tools for generating your Embedded Linux System. The next paragraph explains the main outputs obtained.

3.7. Buildroot Output.

The main output files of the execution of the previous steps can be located in the folder “build/images”. Fig. 3.9 summarizes the use of **Buildroot**. Buildroot generates a bootloader, a kernel image, and a file system.

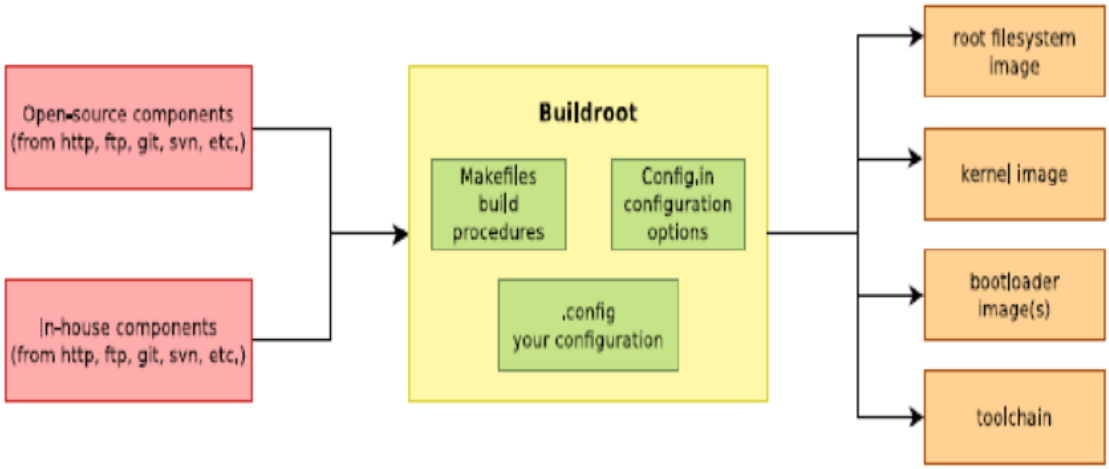


Fig. 3.9 Schematic representation of the Buildroot tool. Buildroot generates the root file system, the kernel image, the bootloader, and the toolchain. Figure copied from "Bootlin" training materials (http://bootlin.com/training/)

In our specific case, the folder content is shown in Fig. 3.10

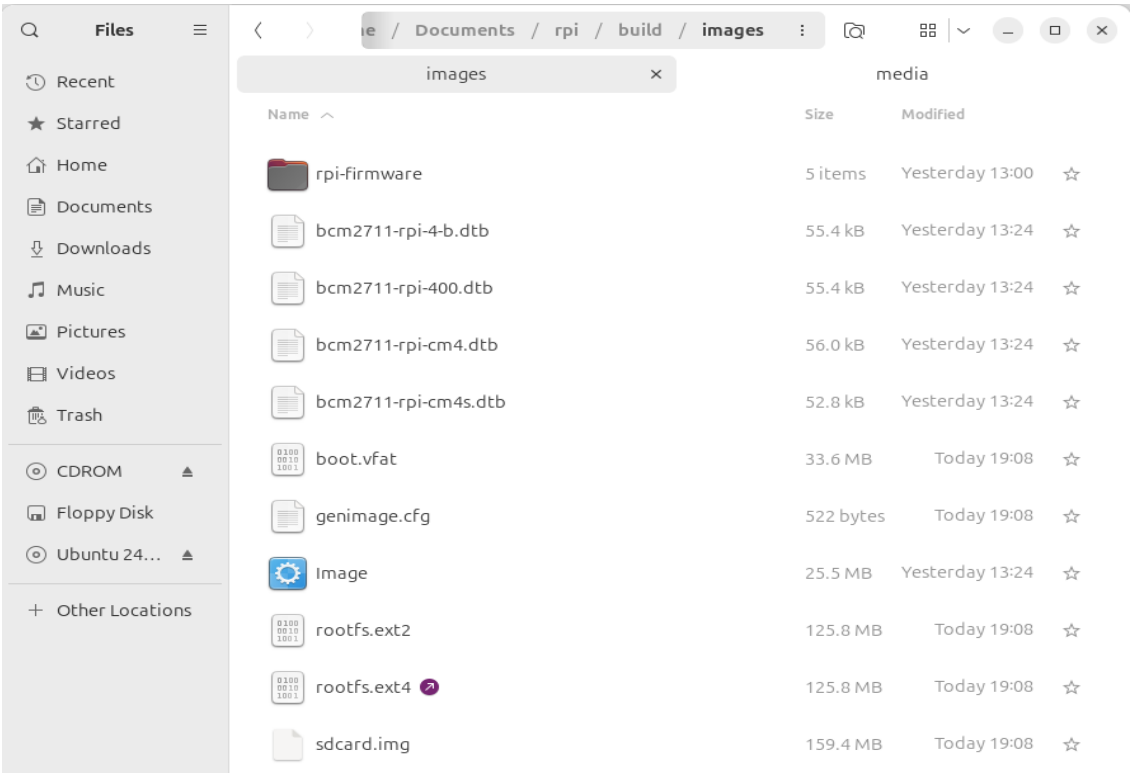


Fig. 3.10 The images folder contains the binary files for our embedded system.

Copy the *sdcard.img* file to your SDcard using this Linux command in the Buildroot folder (*sdb* is typically the device assigned to the sdcard, unless you have other removable devices connected to the system):

```
$ sudo dd if=./images/sdcard.img of=/dev/sd<x> bs=10M
```

Warning

<x> is the identification used by Linux for your microSD card, tipically "b" or "c". **Never** use "a" because this is the operating system hardisk

Remember to format again the microSDcard if you need to repeat this process otherwise you will have errors when Linux is booting.

See also

Linux *gparted* is an excellent tool for partitioning and formatting the SD card.

3.8. Booting the Raspberry Pi.

Fig. 3.11 displays a Raspberry Pi. The description of this card, its functionalities, interfaces, and connectors are explained in the ref [RD2]. The fundamental connection requires:

1. To connect a USB to RS232 adapter (provided) to the raspberry-pi expansion header (see Fig. 3.12, and Fig. 3.13). This adapter provides the serial line interface as a console in the Linux host operating system.
2. To connect the power supply with the micro-USB connector provided (5 v).
3. To connect the Ethernet cable to the RJ45 port if it is available (at home, not the case of UPM's Lab).

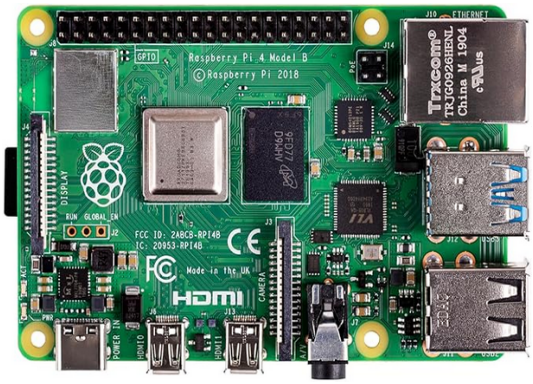


Fig. 3.11 RaspBerry-Pi 4 Model B hardware with main elements identified

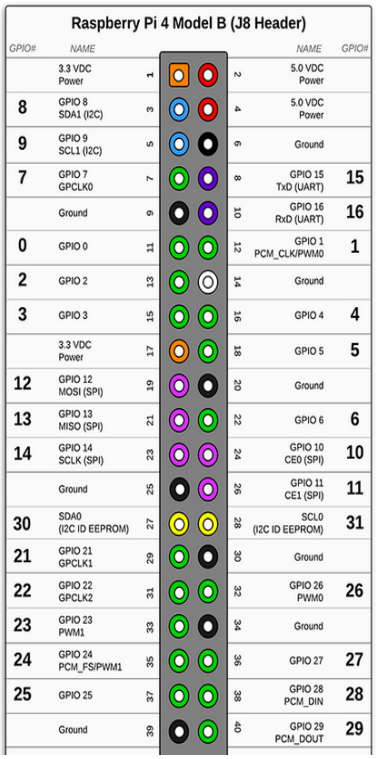


Fig. 3.12 Raspberry-PI 4 header terminal identification.

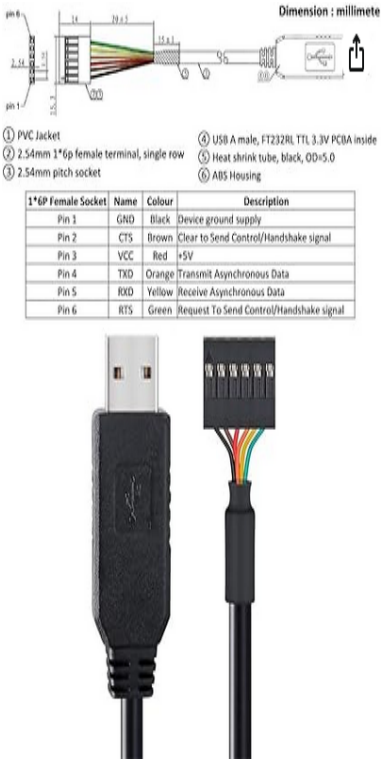


Fig. 3.13 Identification of the terminals in the USB-RS232 adapter

Table 3.9 FDTI Terminals

RPI connector	FDTI Terminal
RXD UART (GPIO16) Pin 10	Terminal 4 (Orange)
TXD UART (GPIO15) Pin 8	Terminal 5 (Yellow)
Ground (GND) Pin 6	Pin 1

The booting process of the Raspberry Pi BCM2711 processor is depicted in Fig. 3.14. Take into account that this System On Chip (SoC), the BCM2711, contains two different processors: a GPU and an ARM CPU. The programs *bootcode.bin* and *start.elf* are written explicitly for the GPU, and the source code is unavailable. Broadcom only provides details of this to customers who sign a commercial agreement. The last executable (*start.elf*) boots the ARM processor and allows the execution of ARM programs such as Linux OS kernel or other binaries such as u-boot bootloader.

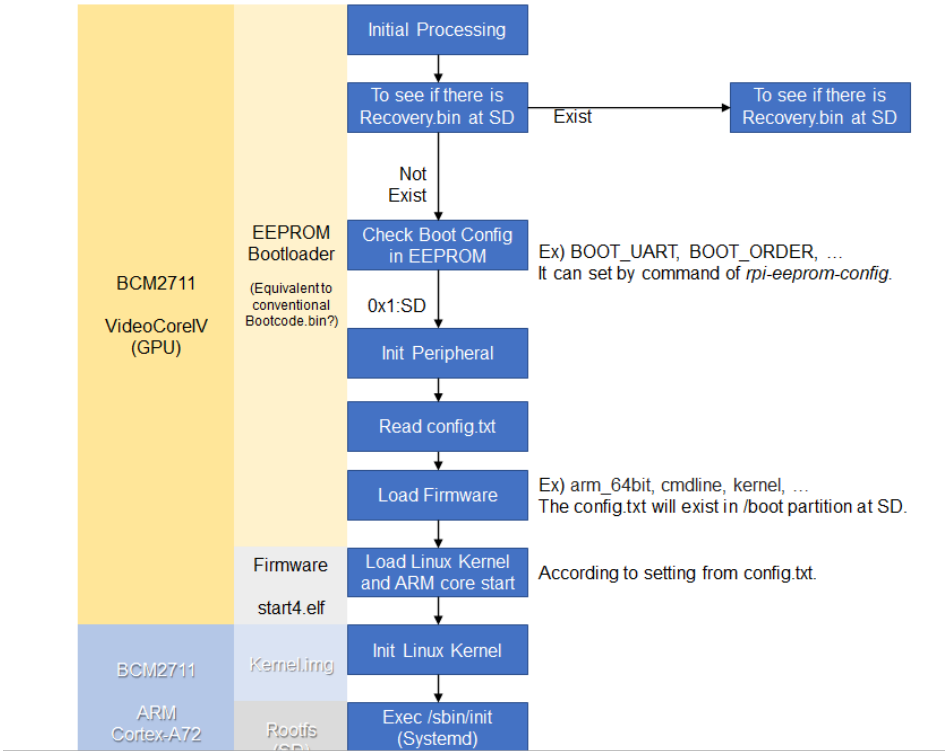


Fig. 3.14 Booting process for BCM2711 processor in the raspberry-pi.

The config.txt file contains essential information to boot the Linux OS and perform the configuration of different hardware elements (look at http://elinux.org/RPiconfig and check the meaning of the different configuration parameters). Verify the content of the config.txt file generated by buildroot and complete it as depicted below (Listing 3.2).

Listing 3.2 config.txt file

```
# Please note that this is only a sample, we recommend you to change
it to fit
# your needs.# You should override this file using
BR2_PACKAGE_RPI_FIRMWARE_CONFIG_FILE.
# See http://buildroot.org/manual.html#rootfs-custom
# and http://elinux.org/RPiconfig for a description of config.txt
syntax

start_file=start4.elf
fixup_file=fixup4.dat

kernel=**Image**

# To use an external initramfs file
#initramfs rootfs.cpio.gz
# Disable overscan assuming the display supports displaying the full
resolution
# If the text shown on the screen disappears off the edge, comment
this out

disable_overscan=1

# How much memory in MB to assign to the GPU on Pi models having
# 256, 512 or 1024 MB total memory
gpu_mem_256=100
gpu_mem_512=100
gpu_mem_1024=100

# Enable UART0 for serial console on ttyAMA0
dtoverlay=miniuart-bt

# enable autoprobing of Bluetooth driver without need of hciattach/
btattach
dtparam=krnbt=on
```

In this example, once the ARM is released from reset, it executes the Image application. This binary application is the Linux Kernel in Image format. The parameters passed to the application specified in the *kernel=<....>* are detailed in the *cmdline.txt* file. For instance, by default, Buildroot generates this one (Listing 3.3):

Listing 3.3 cmdline.txt file

```
root=/dev/mmcblk0p2 rootwait console=tty1 console=ttyAMA0,115200
```

In the Linux machine, open a Terminal and execute the program *sudo putty* with **sudo rights** (sudo putty), in a second a window appears. Configure the parameters using the information displayed

in Fig. 3.15 (for the specific case of *putty*), and then press "Open". **Apply the power to the Raspberry PI,** and you will see the booting messages.

Tip

[Serial interface identification in Linux]: In Linux the serial devices are identified typically with the names /dev/ttyS0, /dev/ttyS1, etc. In the figure, the example has been checked with a serial port implemented with a USB-RS232 converter. This is the reason why the name is **/dev/ttyUSB0**. In your computer, you need to find the identification of your serial port. Use Linux **dmesg** command to do this.

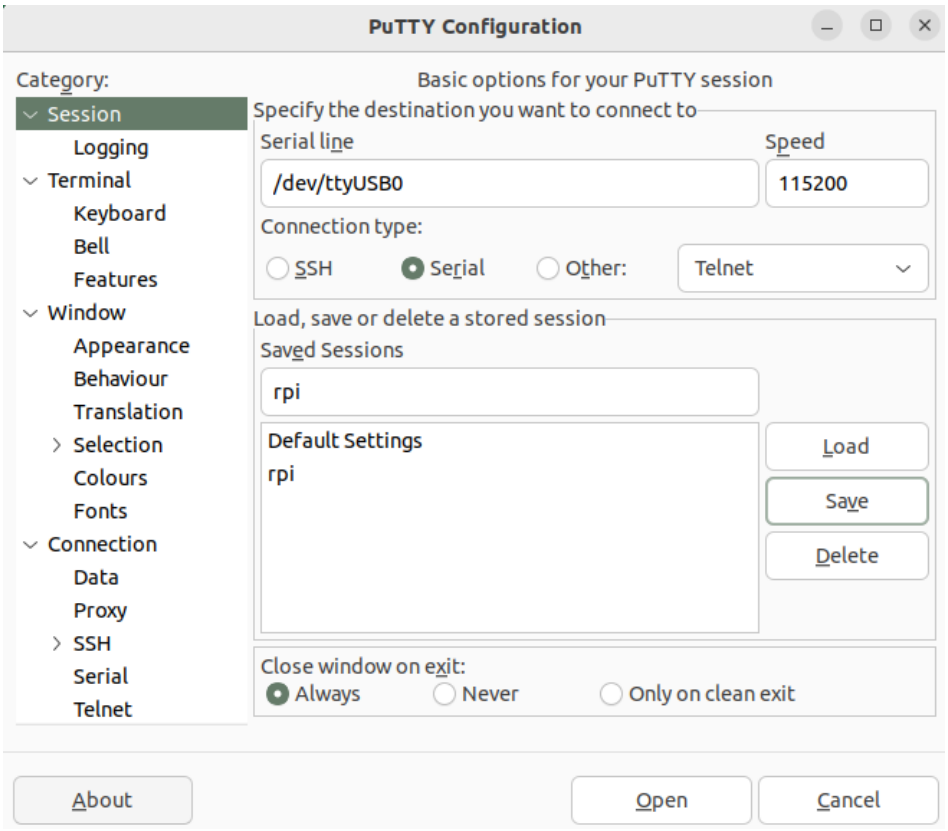


Fig. 3.15 Putty program main window.

After a few seconds, you will see a lot of messages displayed on the terminal. Linux kernel is booting, and the operating system is running its configuration and initial daemons. If the system boots correctly, you will see an output like the one represented in Fig. 3.16. Introduce the username *root* (password in case you have configured it), and the Linux shell will be available for you.

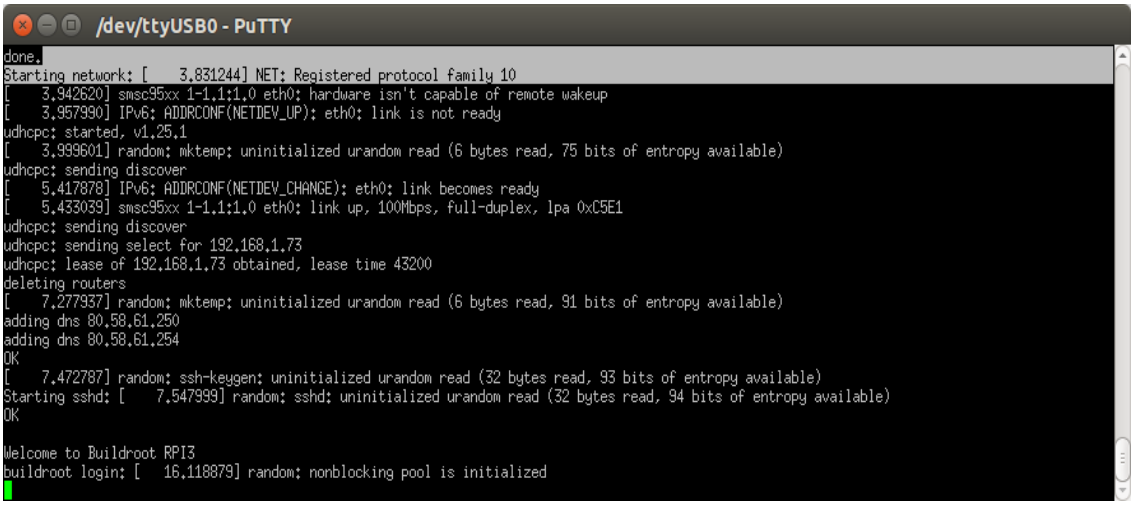


Fig. 3.16 Linux Running in the RaspBerry Pi

Tip

[DHCP Server]: The DHCP server providing the IP address to the RPI should be active in your network. In the UPM ETSIST labs, there is no cabled network, only WIFI. If you are using the RPI at home, the DHCP server is running in your router. The method used to assing IP addresses is different from one manufactures to others. If you want to know the IP address assigned, you have two options: use a serial cable connected to the RPI (*ifconfig* command) or check the router status web page and display the table of the DHCP clients connected. Looking for the MAC in the list, you will obtain the IP address.

3.9. Connecting the RPI to the cabled ethernet network

3.9.1. Inspecting the configuration of the network interface generated automatically by Buildroot

Inspect the content of */etc/network/interfaces* and */etc/init.d/S40network*. You will see content similar to this in the *interfaces* file:

```
# interface file auto-generated by buildroot

auto lo
iface lo inet loopback

auto eth0
```

```
iface eth0 inet dhcp
     pre-up /etc/network/nfs_check
     wait-delay 15
     hostname $(hostname)
```

This configuration activates the use of eth0 with DHCP support. Test the connectivity, trying to connect to another computer in the laboratory. Use the ping command.

Note

[Help]: If you run the ping command in the Raspberry trying to connect with a computer in the laboratory, you probably obtain a connection timeout. Consider that computers running Windows could have the firewall activated. You can also try to run the ping on a windows computer or on Linux virtual machine. In this case, the RPI does not have a firewall running, and the connection should be successful.

Question

What is the MAC address of your RPI interface? Use the *dmesg* command to see the kernel boot parameters and identify the method used to get the MAC address from the hardware.

3.10. Adding WIFI support

3.10.1. Adding mdev support to Embedded Linux

The folder <buildroot-folder>*/package/busybox* contains two files named S10mdev and mdev.conf. These files have to be added to the target filesystem. This step is done by adding these commands to the *<buildroot-folder>/board/raspberrypi4-64/post-build.sh* script:

```
cp <buildroot-folder>/package/busybox/S10mdev ${TARGET_DIR}/etc/
init.d/S10mdev
chmod 755 ${TARGET_DIR}/etc/init.d/S10mdev
cp <buildroot-folder>/package/busybox/mdev.conf ${TARGET_DIR}/etc/
mdev.conf
```

Note

[mdev] mdev provides a method to add or remove hotplug devices in Linux.

3.10.2. Adding the Broadcom firmware support for Wireless hardware

The hardware element included in the RPI-4 for the Wireless communication is implemented with the BCM43438 chip. It is needed to include the software packages with the firmware's chip and the wireless utilities.

1. Execute "make ..... menuconfig". Navigate to "Target Packages->Hardware Handling->Firmware-> bcrmfmac-sdio-firmware-rpi" and select the "bcrmfmac-sdio-firmware-rpi-wifi".
2. Before compiling Buildroot we need to add more software supporting the configuration of the WIFI.
    1. Navigate to "Target Packages->Networking Applications" and select
        - "crda"
        - "ifupdown scripts"
        - "iw"
        - "wireless-regdb"
        - "wireless tools"
        - "wpa_supplicant"
            1. "Enable EAP"
            2. "Enable WPS"
            3. "Install wpa_cli binary"
            4. "Install wpa_client shared library"
            5. "Instal wpa_passphrase binary"
    2. Add these lines to ./board/rapsberrypi4-64/post-build.sh.

```
cp <buildroot-folder>/board/raspberrypi4-64/interfaces ${TARGET_DIR}/
etc/network/interfaces
```

```
cp <buildroot-folder>/board/raspberrypi4-64/wpa_supplicant.conf ${TARGET_DIR}/etc/wpa_supplicant.conf
```

1. Create the file **<buildroot-folder>*/board/raspberrypi4-64/interfaces* adding the highlighted content:

```
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp
     pre-up /etc/network/nfs_check
     wait-delay 15
     hostname $(hostname)

auto wlan0
iface wlan0 inet dhcp
       pre-up wpa_supplicant -B -iwlan0 -c/etc/wpa_supplicant.conf
       post-down killall -q wpa_supplicant
       wait-delay 15
```

1. Create the file **<buildroot-folder>*/board/raspberrypi4-64/wpa_supplicant.conf* with this content (ask professors about the values to be provided as SSID and Key-passwd). You can define as many WIFIs as you want.

```
network={
    ssid="SSID"
    key_mgmt=WPA-PSK
    psk="PASSWORD"
    priority=9
}
```

1. Perform a **make** and burn again the new image in the micro SDcard. Boot the Raspberry and check that you can connect to the wireless network.

3.11. Customizing the Linux kernel

3.11.1. Kernel configuration

Execute the following command to configure the Linux Kernel in Buildroot

```
$ make  O=$PWD -C /home/ubuntu/Documents/rpi/buildroot-2025.02.9/
linux-menuconfig
```

Using the different menus you can configure specific kernel features, add support for specific device drivers and multiple additional functionalities. The compilation of the Linux kernel package is done with this command:

```
$ make  O=$PWD -C /home/ubuntu/Documents/rpi/buildroot-2025.02.9/
linux-rebuild
```

In order to get the new sdcard.img file execute:

```
$ make  O=$PWD -C /home/ubuntu/Documents/rpi/
buildroot-2025.02.9/
```

4. Using the integrated development environment Eclipse/CDT

4.1. Eclipse IDE for C/C++ developers

The Eclipse IDE CDT is installed in the virtual machine. You can execute it running eclipse in a window terminal.

4.2. Cross-Compiling applications using Eclipse

How will a program be compiled and linked? Remember that we are developing cross applications. We are developing and compiling the code in a Linux x86_64 architecture, and we are executing it on an ARM one (see Fig. 4.1).

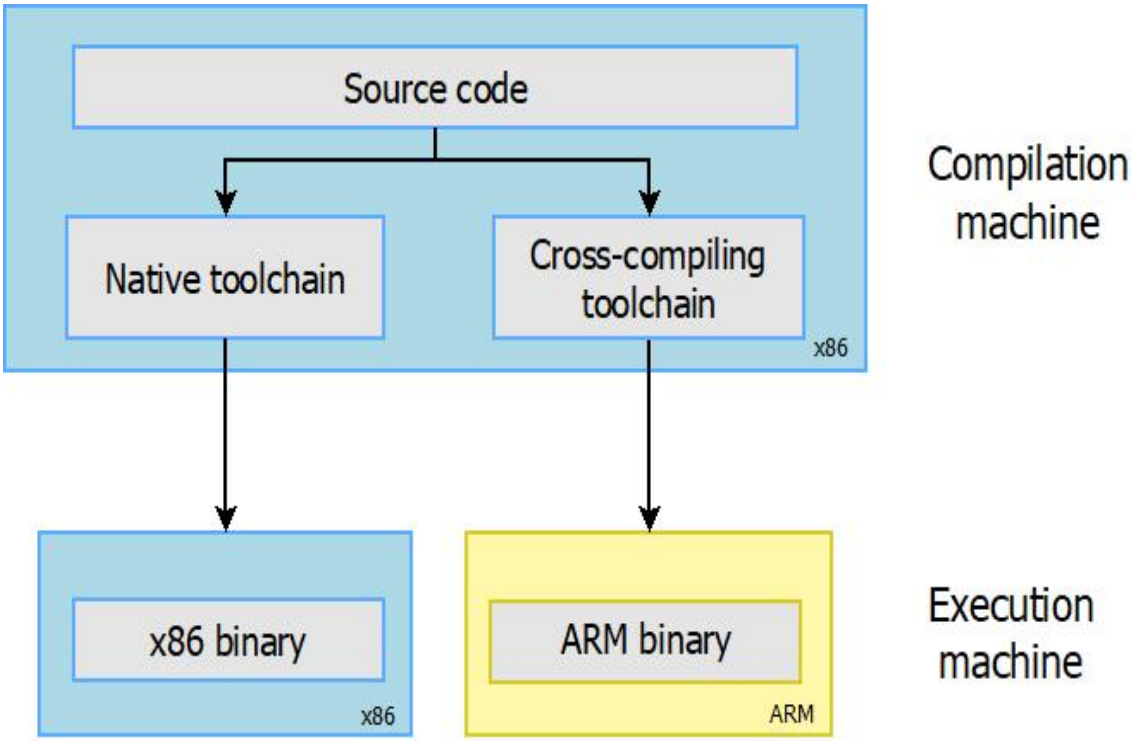


Fig. 4.1 Cross Toolchain. Figure copied from “Free Electrons” training materials (http://free-electrons.com/training/)

The first question is where the cross-compiler and other cross-tools are located. The answer is this: in the folder *build/host/usr/bin*. If you inspect this folder’s content, you can see the entire

compiling, linking, and debugging tools (see Fig. 4.2). These programs are executed in your x86_64 computer, but they generate code for the ARM processor.

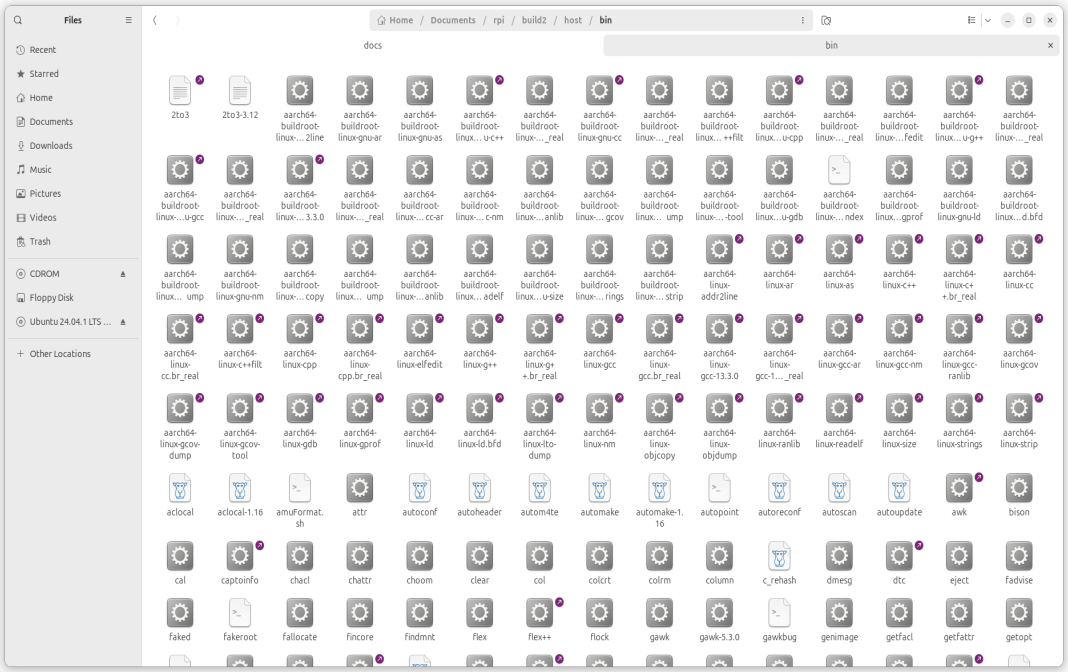


Fig. 4.2 Cross-compiling tools installed in the host computer

In a Terminal window execute the following commands:

```
# change the directory to the folder where `build` directory is
$ cd ./build/host
$ source environment-setup
# run eclipse in the same terminal. In this case eclipse is available
in your ubuntu user folder
$ /home/ubuntu/eclipse/cpp-2024-09/eclipse/eclipse &
```

The **environment-setup** file contains the code listed below.

```
cat <<'EOF'
 _                 _ _       _                          _
| |__   _   _(_) | __| |_ __ ___   ___ | |_
| '_ \| | | | | |/ _` | '__/ _ \ / _ \| __|
| |_) | |_| | | | (_| | | | (_) | (_) | |_
|_.__/ \__,_|_|_|\__,_|_|  \___/ \___/ \__|

       Making embedded Linux easy!

Some tips:
* PATH now contains the SDK utilities
* Standard autotools variables (CC, LD, CFLAGS) are exported
* Kernel compilation variables (ARCH, CROSS_COMPILE, KERNELDIR) are
```

```
exported
* To configure do "./configure $CONFIGURE_FLAGS" or use
  the "configure" alias
* To build CMake-based projects, use the "cmake" alias

EOF
if [ x"$BASH_VERSION" != x"" ] ; then
     SDK_PATH=$(dirname $(realpath "${BASH_SOURCE[0]}"))
elif [ x"$ZSH_VERSION" != x"" ] ; then
     SDK_PATH=$(dirname $(realpath $0))
else
     echo "unsupported shell"
fi
export "AR=aarch64-buildroot-linux-gnu-gcc-ar"
export "AS=aarch64-buildroot-linux-gnu-as"
export "LD=aarch64-buildroot-linux-gnu-ld"
export "NM=aarch64-buildroot-linux-gnu-gcc-nm"
export "CC=aarch64-buildroot-linux-gnu-gcc"
export "GCC=aarch64-buildroot-linux-gnu-gcc"
export "CPP=aarch64-buildroot-linux-gnu-cpp"
export "CXX=aarch64-buildroot-linux-gnu-g++"
export "FC=aarch64-buildroot-linux-gnu-gfortran"
export "F77=aarch64-buildroot-linux-gnu-gfortran"
export "RANLIB=aarch64-buildroot-linux-gnu-gcc-ranlib"
export "READELF=aarch64-buildroot-linux-gnu-readelf"
export "STRIP=aarch64-buildroot-linux-gnu-strip"
export "OBJCOPY=aarch64-buildroot-linux-gnu-objcopy"
export "OBJDUMP=aarch64-buildroot-linux-gnu-objdump"
export "AR_FOR_BUILD=/usr/bin/ar"
export "AS_FOR_BUILD=/usr/bin/as"
export "CC_FOR_BUILD=/usr/bin/gcc"
export "GCC_FOR_BUILD=/usr/bin/gcc"
export "CXX_FOR_BUILD=/usr/bin/g++"
export "LD_FOR_BUILD=/usr/bin/ld"
export "CPPFLAGS_FOR_BUILD=-I$SDK_PATH/include"
export "CFLAGS_FOR_BUILD=-O2 -I$SDK_PATH/include"
export "CXXFLAGS_FOR_BUILD=-O2 -I$SDK_PATH/include"
export "LDFLAGS_FOR_BUILD=-L$SDK_PATH/lib -Wl,-rpath,$SDK_PATH/lib"
export "FCFLAGS_FOR_BUILD="
export "DEFAULT_ASSEMBLER=aarch64-buildroot-linux-gnu-as"
export "DEFAULT_LINKER=aarch64-buildroot-linux-gnu-ld"
export "CPPFLAGS=-D_LARGEFILE_SOURCE -D_LARGEFILE64_SOURCE -
D_FILE_OFFSET_BITS=64"
export "CFLAGS=-D_LARGEFILE_SOURCE -D_LARGEFILE64_SOURCE -
D_FILE_OFFSET_BITS=64  -Os -g0 -D_FORTIFY_SOURCE=1"
export "CXXFLAGS=-D_LARGEFILE_SOURCE -D_LARGEFILE64_SOURCE -
D_FILE_OFFSET_BITS=64  -Os -g0 -D_FORTIFY_SOURCE=1"
export "LDFLAGS="
export "FCFLAGS= -Os -g0"
export "FFLAGS= -Os -g0"
export "PKG_CONFIG=pkg-config"
export "STAGING_DIR=$SDK_PATH/aarch64-buildroot-linux-gnu/sysroot"
```

```
export "INTLTOOL_PERL=/usr/bin/perl"
export "ARCH=arm64"
export "CROSS_COMPILE=aarch64-buildroot-linux-gnu-"
export "CONFIGURE_FLAGS=--target=aarch64-buildroot-linux-gnu --
host=aarch64-buildroot-linux-gnu --build=x86_64-pc-linux-gnu --
prefix=/usr --exec-prefix=/usr --sysconfdir=/etc --localstatedir=/var
--program-prefix="
alias configure="./configure ${CONFIGURE_FLAGS}"
alias cmake="cmake -DCMAKE_TOOLCHAIN_FILE=$SDK_PATH/share/buildroot/
toolchainfile.cmake -DCMAKE_INSTALL_PREFIX=/usr"
export "PATH=$SDK_PATH/bin:$SDK_PATH/sbin:$PATH"
export "KERNELDIR=/home/ubuntu/Documents/rpi/build/build/linux-
custom/"
```

This script when is sourced in a terminal window sets all the environment variables needed to use the cross-compilation tools and adds the folder of cross-tools to the Linux *PATH* variable.

The execution of eclipse popups a window inviting you to enter the workspace (see Fig. 4.3). The workspace is the folder that contain eclipse projects created by the user. You can have as many workspaces as you want. Please specify a folder in your account.

Tip

The figures displayed in the following paragraphs can be different depending on the Eclipse version installed

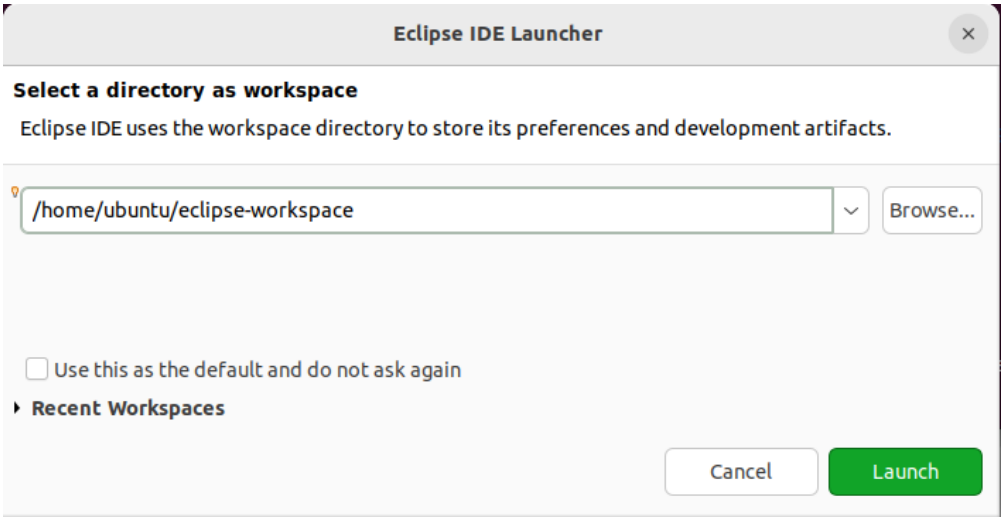


Fig. 4.3 Selection of the workspace for Eclipse. Use a folder in your account.

Select Ok, and the welcome window of Eclipse will be shown (see Fig. 4.4).Next, close the welcome window and the main eclipse window will be displayed (see Fig. 4.5).

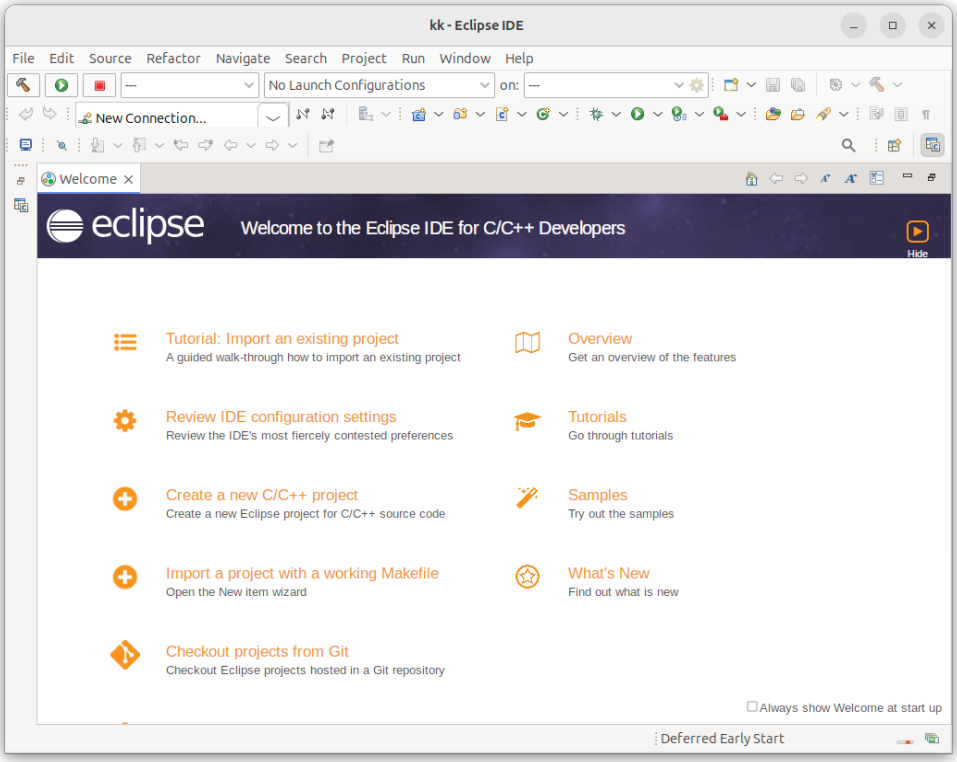


Fig. 4.4 Eclipse welcome window.

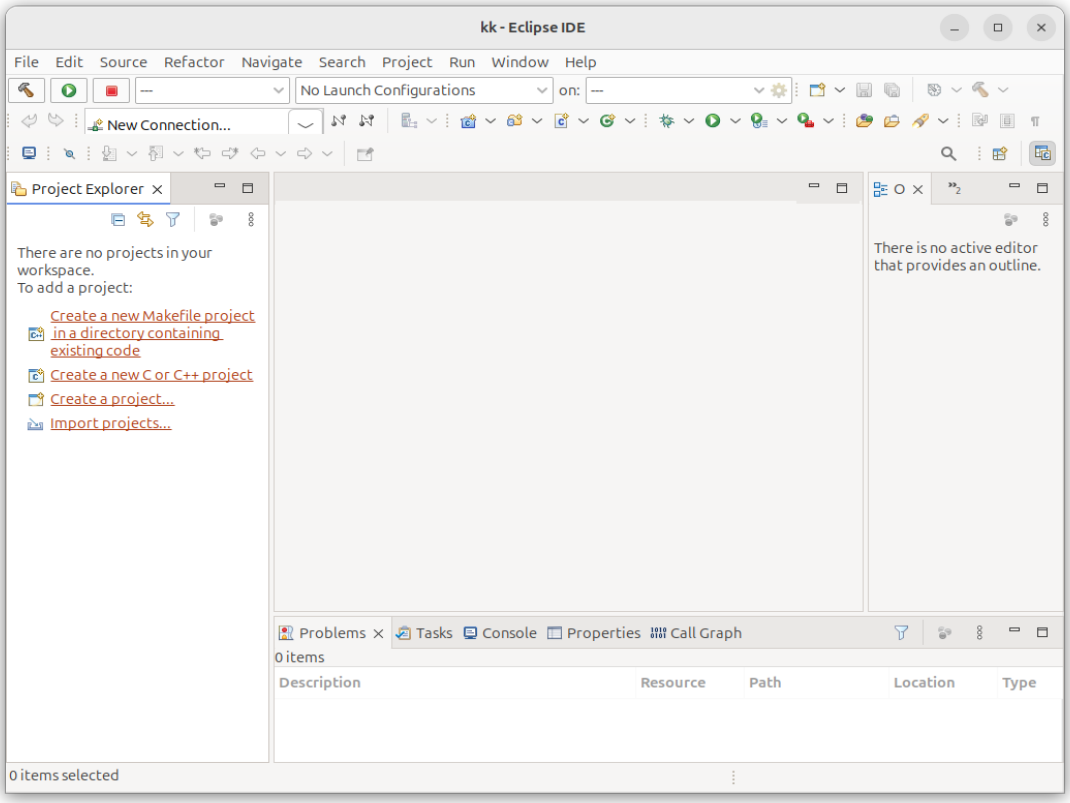


Fig. 4.5 Eclipse main window.

In a terminal window create an empty folder. In this folder create the following files with the content described in Listing 4.1, Listing 4.2 , Listing 4.3, and Listing 4.4. The Makefile uses the environment variables that are defined in the environment where the *makefile* is run.

Listing 4.1 Makefile

```
LIBS= -lpthread -lm #Libraries used if needed
SRCS= main.cpp func.cpp
BIN=app
CFLAGS+= -g -O0
OBJS=$(subst .cpp,.o,$(SRCS))
all : $(BIN)
$(BIN): $(OBJS)
    @echo [link] $@
    $(CXX) -o $@ $(OBJS) $(LDFLAGS) $(LIBS)
%.o: %.cpp
    @echo [Compile] $<
    $(CXX) -c $(CFLAGS) $< -o $@
clean:
    @rm -f $(OBJS) $(BIN)
```

Listing 4.2 main.cpp

```
#include "func.h"
#include <iostream>
int main(void){
    int b=2;
    std::cout<<"A is: "<< fun(b) << std::endl;
}
```

Listing 4.3 func.h

```
#ifndef __FUNC_H
#define __FUNC_H
    int fun(int);
#endif
```

Listing 4.4 func.cpp

```
int fun(int b){
    int a=b*2;
    return a;
}
```

In Eclipse select in the left part of the windows *Import projects*. A new window is popup, select then *C/C++* and the option *Existing Code as Makefile Project*. The window shown in Fig. 4.6 is displayed. Complete the name of the project, select the folder with the code and check *Cross GCC in Toolchain for Indexer Settings*.

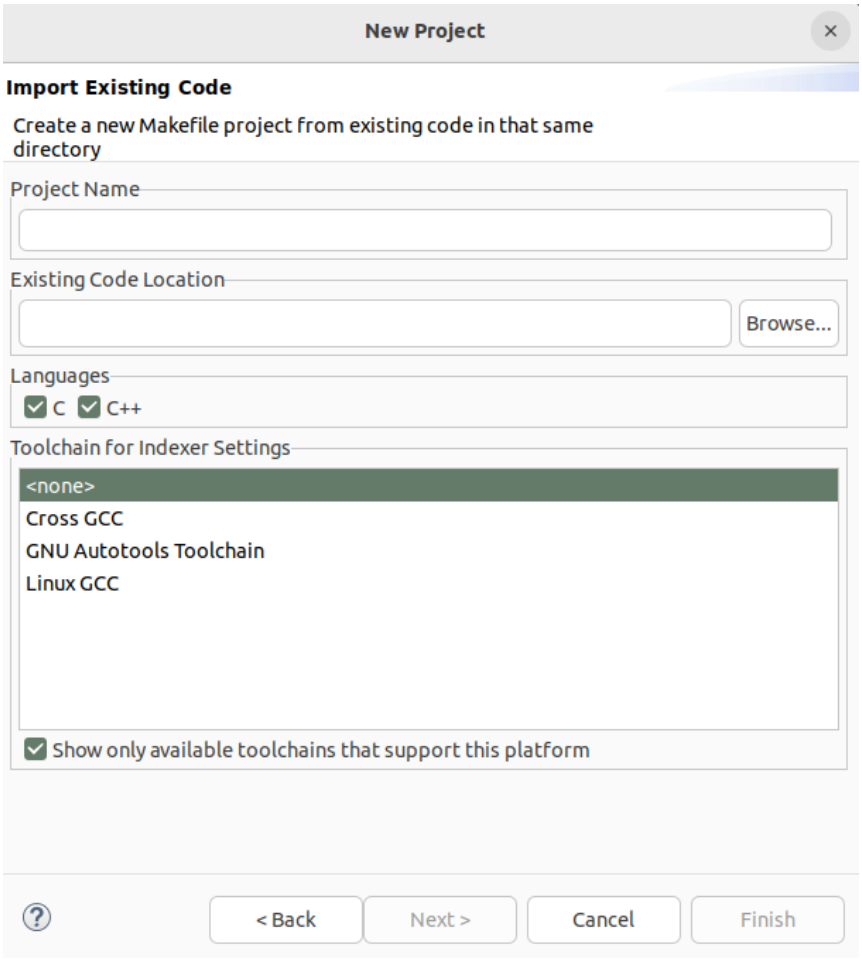


Fig. 4.6 Importing the code.

4.3. Building a project

Once you have configured the cross-chain in Eclipse you can build your project using Project->Build Project. If everything is correct, you will see the eclipse project as represented in Fig. 4.7 . You can clean the project (remove the executable and objects) with *Clean*.

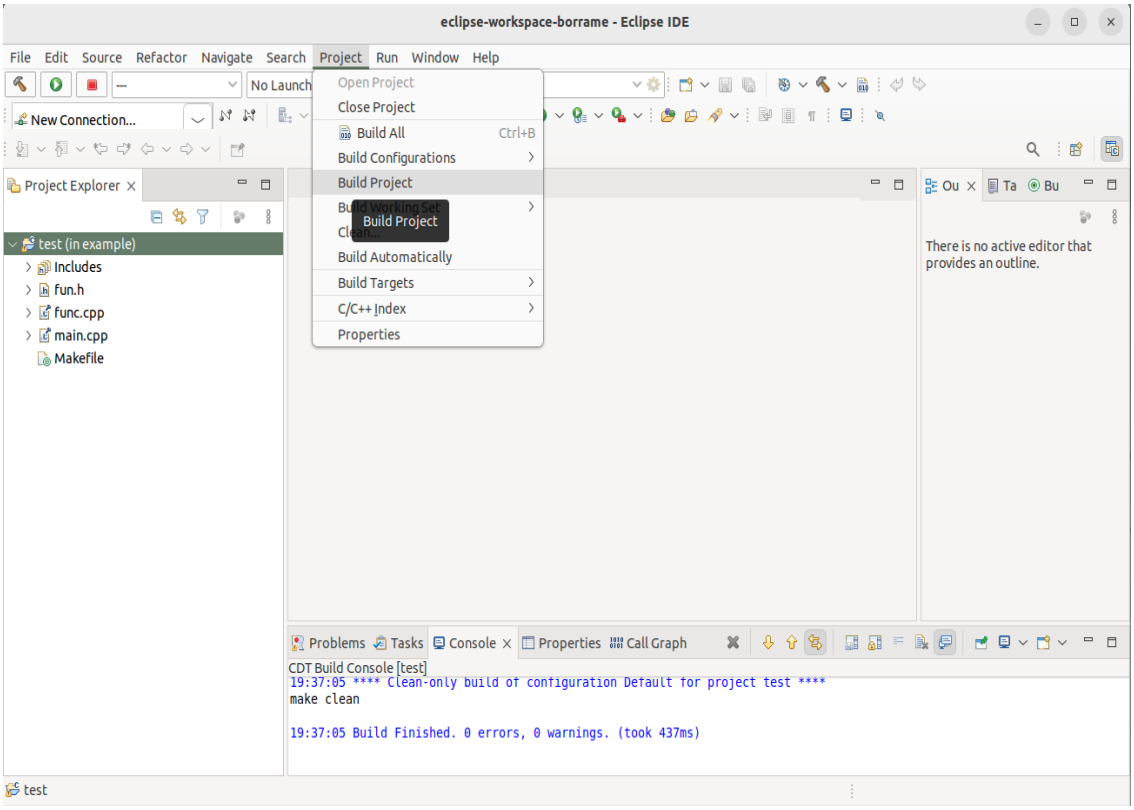


Fig. 4.7 Eclipse project compiled (Binaries has been generated)

Note

[Console in Eclipse]: Have a look at the messages displayed in the Console. You will see how eclipse is calling the cross compiler with different parameters.

4.4. Moving the binary to the target

In order to copy the executable to the target, you have different options. You can use the Linux application called *scp* or other similar applications. In our case, we are going to use the "Other Locations...." utility included in the nautilus explorer (Fig. 4.8). Specify in Server Address *ssh://<ip address>*

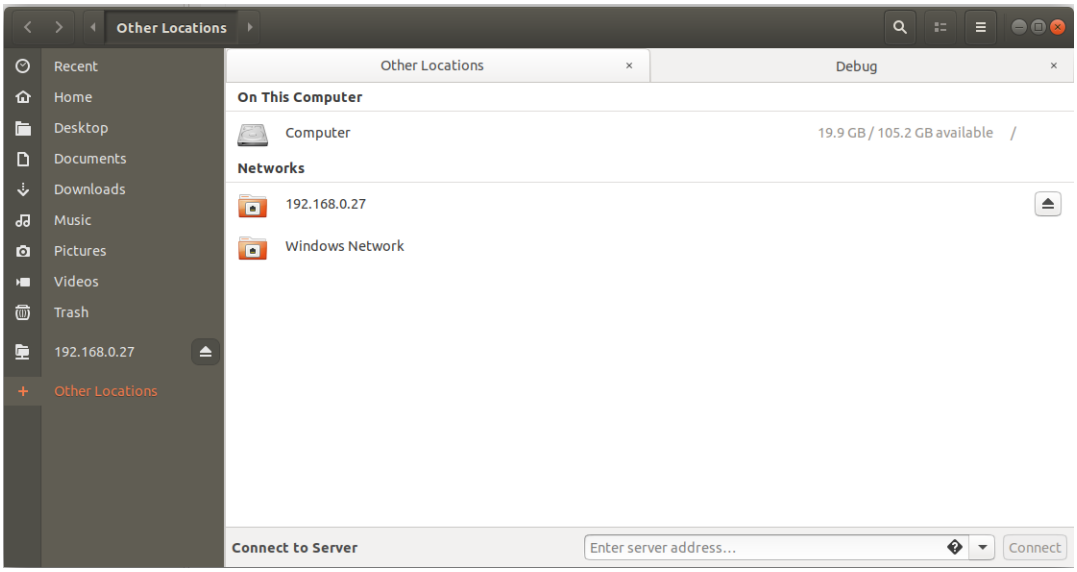


Fig. 4.8 Connect to Server" option in Nautilus explorer

4.5. Executing the application

You can run the Raspberry PI program using putty (remember that once you have a network connection available in the RPI you can also use putty to connect to it).

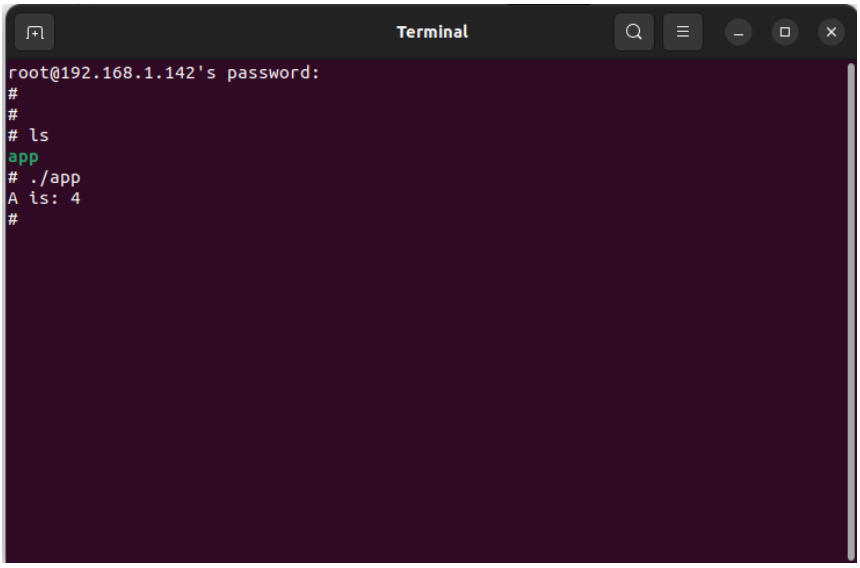


Fig. 4.9 Run test program in Raspberry Pi

⚠ Warning

Warning. If you experiment problems using ssh, delete the *.ssh* folder in your home directory.

4.6. Automatic debugging using gdb and gdbserver

You can directly debug the program running in the RPI using Eclipse. There are two methods to do it: manually and automatically. In the manual method, firstly, you need to copy the executable program to the RPI, change the file permissions to “executable” and execute the program to be debugged using *gdbserver* utility. Of course, this is a time-consuming process and very inefficient. The alternative solution is to use automatic debugging. In order to debug your applications, we need to define a debug session and configure it. Firstly, *Select Run->Debug Configurations* and generate a new configuration under *C/C++ Remote Application*. You need to complete the different tabs available in this window. The first one is the main tab (see Fig. 33). You need to configure here the path to the C/C++ application to be debugged, the project name, the connection with the target (you will need to create a new one using the IP address of your RPI), the remote path where your executable file will be downloaded, and the mode for the debugging (Automatic Remote Debugging Launcher). Secondly, in the argument tab, you can specify the arguments of your executable program. It is very important here that you can also specify the working directory path where the executable will be copied and launched (you need to have rights in this folder).

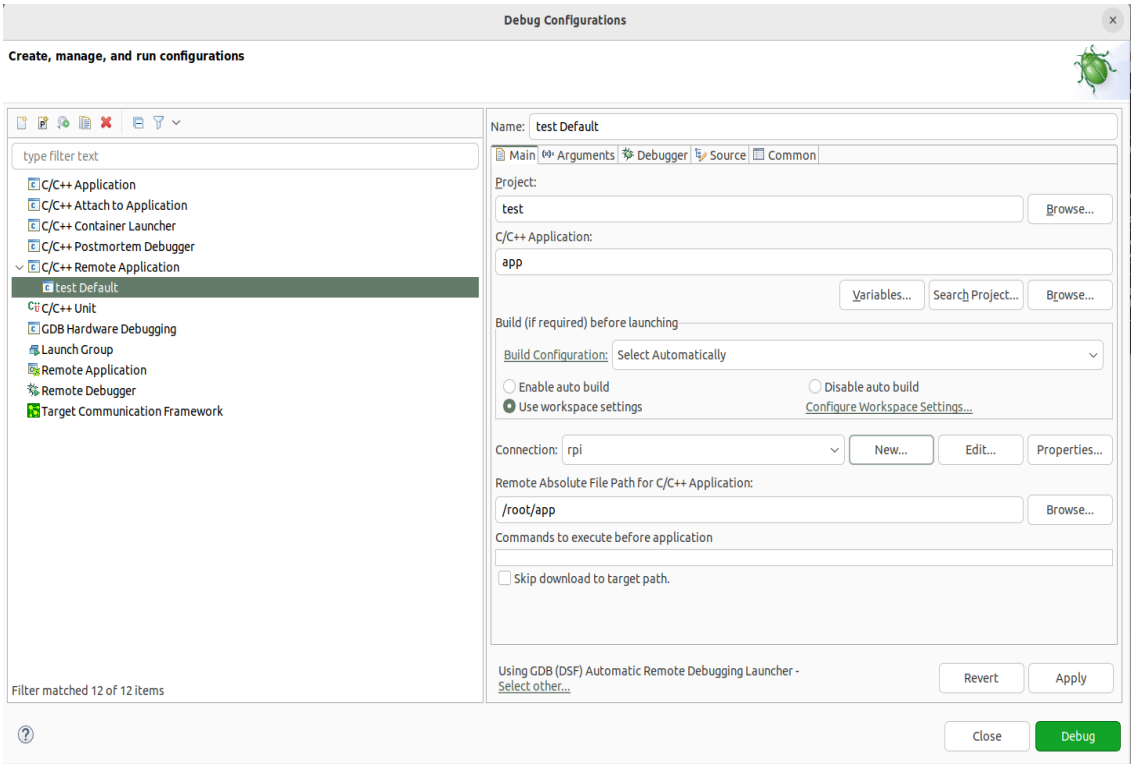


Fig. 4.10 Creating a Debug Configuration

In the debugger window you need to configure the path of your cross gdb application. Remember that we are working with a cross-compiler, cross debugging. Therefore, you need to provide here the correct path of your gdb. The GDB command file (.gdbinit) must be specified, providing a path with an empty file. In the Gdbserver settings tab, you need to provide the path to the gdbserver in the target and the TCP/IP port used (by default 2345).

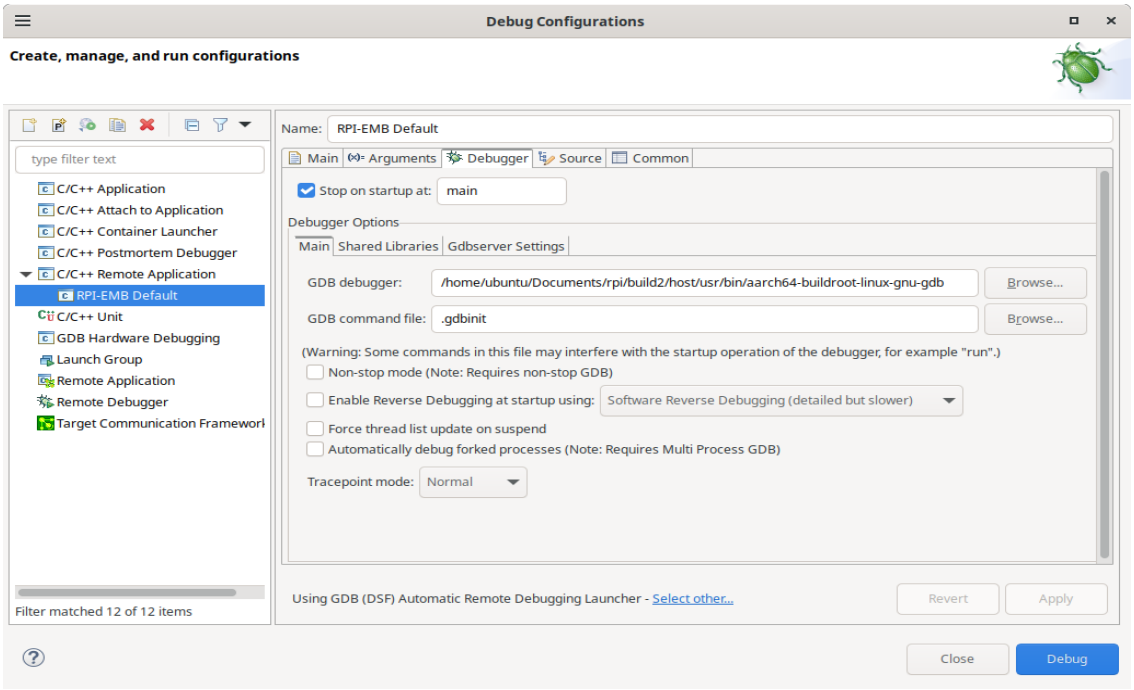


Fig. 4.11 Debug configuration, including the path to locate the cross gdb tool.

Now, press Debug in Eclipse window, and you can debug your application remotely.

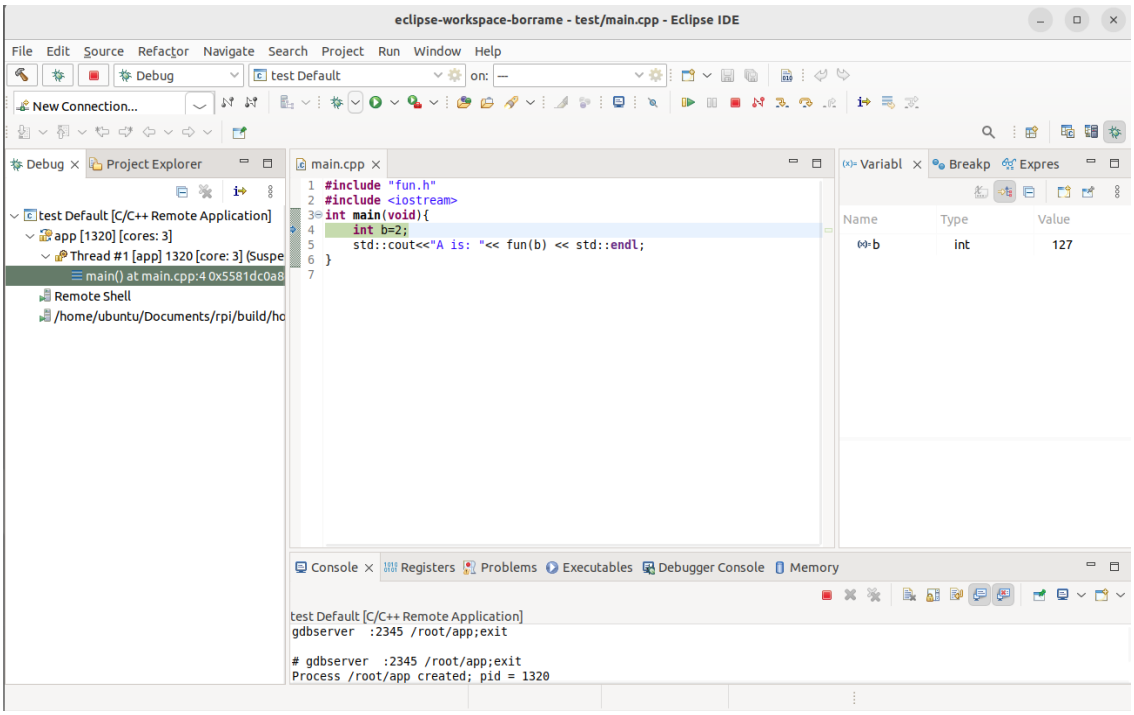


Fig. 4.12 Debugging session on the RPI remotely

5. Preparing the VMware Ubuntu Virtual Machine.

5.1. Download VMware Workstation Player.

The software can be downloaded from this link. Follow the instructions to set up the application on your computer.

5.2. Installing Ubuntu 24.04.01 LTS as a virtual machine.

Note

It is mandatory to install Ubuntu 24.04.01 LTS version. Use this link to download it http://ftp.rediris.es/sites/releases.ubuntu.com/noble

The first step is to download Ubuntu. You will download an ISO image with this Linux operating System. Run WMware player and install Ubuntu using the VMWare player instructions. Select the *Typical* configuration and reserve at least 100G/150G for the hard disk. In the hardware configuration select 8GByte of RAM, if possible 4 processors, and the USB 3.1 support. The installation time will be half an hour, more or less, depending on your computer. Moving a virtual machine from one computer to another is a time-consuming task; therefore, take this into account to minimize the development time.

5.3. Installing synaptic

If you need to install software packages, you can do it using the linux terminal command *apt-get*. Another alternative process is the use of the synaptic utility. In order to use it, you need to install it using this command:

```
$ sudo apt-get install synaptic
```

Once installed, you can search and execute the synaptic program. When you click two times over the package, it will show all the dependent packages that would be installed (see Fig. 5.1).

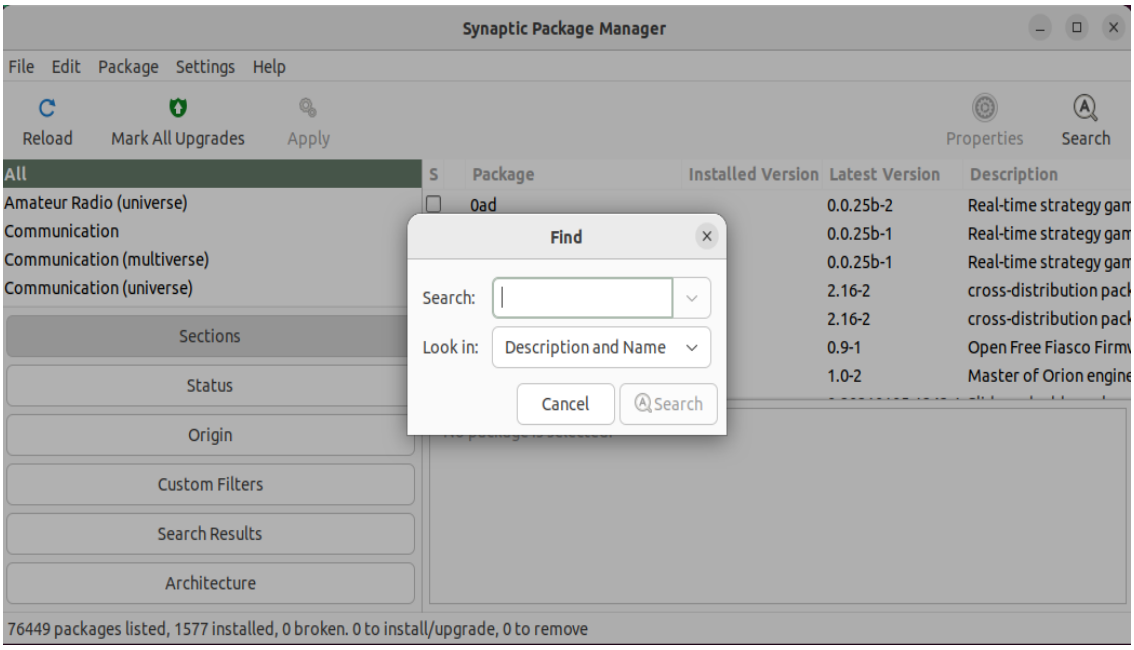


Fig. 5.1 Synaptic window

5.4. Installing putty

You need to execute:

- sudo apt-get install putty

5.5. Installing packages for supporting Buildroot.

Using buildroot requires some software packages that have to be installed in the VM. These are listed in this link http://buildroot.uclibc.org/downloads/manual/manual.html#requirement. You need to install at least:

- build-essential
- git
- libncurses-dev

5.6. Installing packages supporting Eclipse

Follow the instructions of this link to install Eclipse. Use the Eclipse C/C++ or the Eclipse for Embedded C/C++ developers.

6. Using Visual Studio Code for Raspberry Pi Development

VSCode is one of the most popular code editors available today. This guide will help you set up VSCode for developing applications for Raspberry Pi. First, we will setup the IDE for remote development on the Raspberry Pi, and then we will configure it for cross-compiling applications for our custom embedded Linux system built with Buildroot.

6.1. Installing VSCode in Ubuntu

First, we need to install VSCode on the host machine, that in our case is an Ubuntu virtual machine. You can get the latest version of VSCode from the official website. Select the **.deb package** (see Fig. 6.1) for Debian-based distributions and once it is downloaded open a terminal in the download folder and run the following command:

```
sudo apt install ./code_*.deb
```

Select **Yes** when prompted to add the Microsoft repository

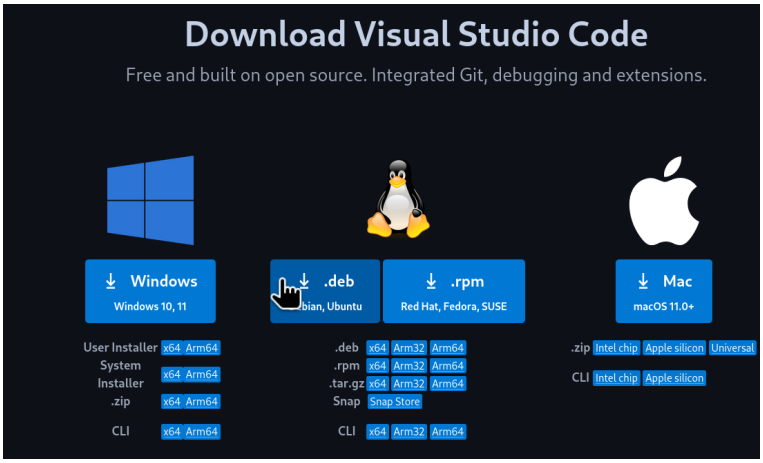


Fig. 6.1 VSCode .deb download

Note

You can also install VSCode using the snap package manager with the following command:

```
sudo snap install --classic code
```

After the installation is completed, open a terminal and launch VSCode to check that it has been installed correctly:

```
code
```

A window should pop up with the VSCode interface (see Fig. 6.2).

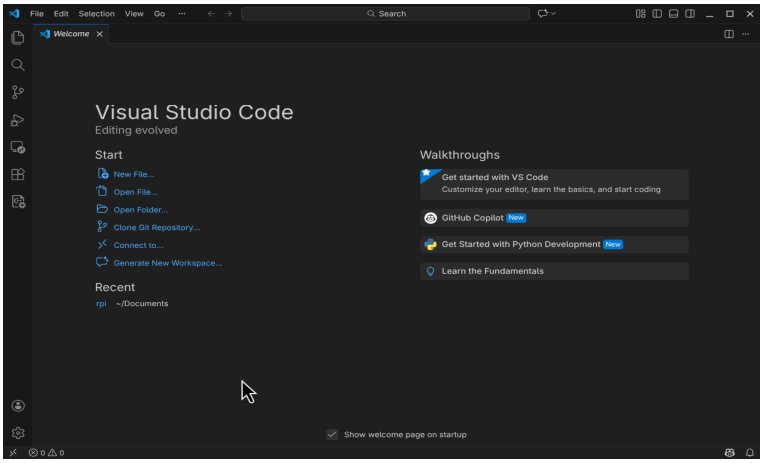


Fig. 6.2 VSCode interface

6.2. Setting up VSCode for Remote Development

Now we are going to set up VSCode for remote development on the Raspberry Pi. In this case we will be able to edit, compile and debug applications on the Raspberry Pi from the VSCode running on our host machine (Ubuntu Virtual Machine). In this use case, **we will not be cross-compiling applications**, but compiling then natively on the Raspberry Pi. This is possible because the Raspberry Pi OS that we installed on the Pi includes all the necessary development tools.

6.2.1. Installing the Remote - SSH Extension

To enable remote development using VSCode, we need to install the *Remote - SSH* extension. To do this, open VSCode and on the left bar, click on the *Extensions* icon. Type "*remote ssh*" in the search bar and install the extension named *Remote - SSH* by Microsoft (see Fig. 6.3).

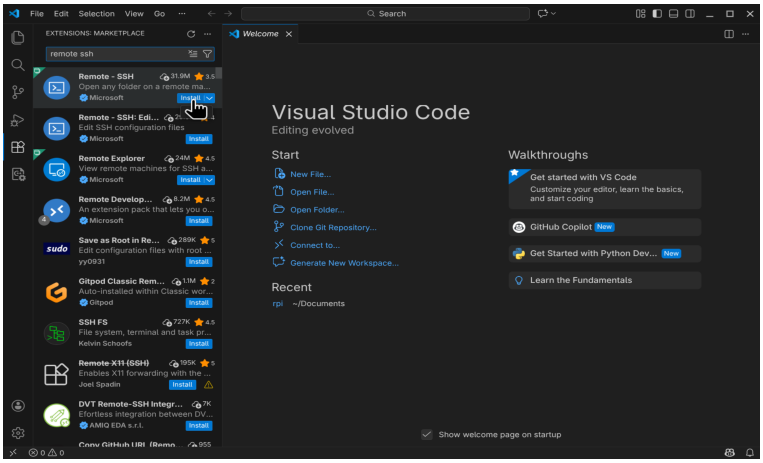


Fig. 6.3 Remote - SSH extension installation

This extension allows us to open remote folders and files on any remote machine with SSH access and work with them just as if they were on our local machine.

6.2.2. Adding the Raspberry Pi as a Remote Host

Now we need to add our Raspberry Pi as a remote host in VSCode. To do this, select click on the bottom left corner icon (see Fig. 6.4) and on the upper menu select *Connect to Host...*.

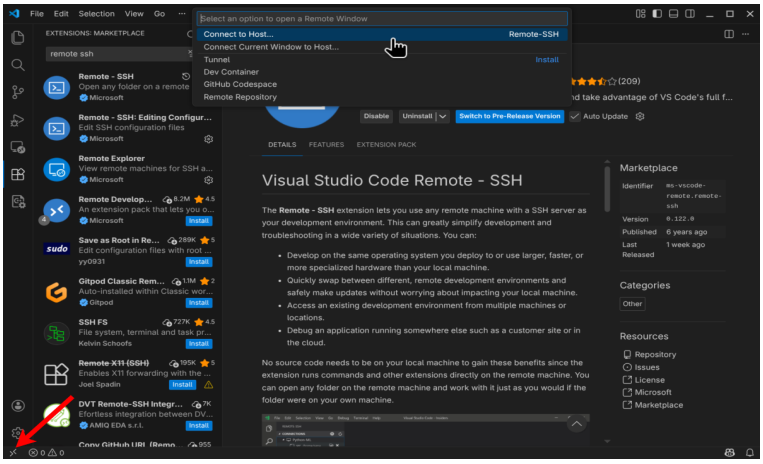


Fig. 6.4 Connect to Host option

Next, select *Add New SSH Host...* and add the SSH connection command to connect to the Raspberry Pi.

```
ssh <username>@<raspberry_pi_ip_address>
```

Replace `<username>` with the username (**rpi-student**) you use to log in to the Raspberry Pi and `<raspberry_pi_ip_address>` with the IP address of your Raspberry Pi. After adding the SSH command, you will be prompted to select the SSH configuration file to update. Select the default

option (*~/.ssh/config*). This will add the Raspberry Pi connection details to the SSH configuration file.

6.2.3. Connecting to the Raspberry Pi

Now, to connect to the Raspberry Pi, click again on the bottom left corner icon and select *Connect to Host...*. This time select the connection that you just added (it should be named with your Raspberry Pi IP address). This will open a new VSCode window and you will be connected to the Raspberry Pi as shown in the bottom left (see Fig. 6.5).

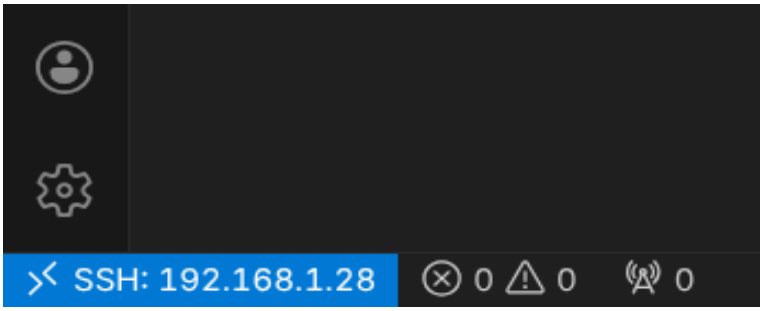


Fig. 6.5 VSCode connected to Raspberry Pi

Now you can open folders and files that are on the Raspberry Pi and edit them directly from VSCode running on your host machine. Take your time and explore the different features of VSCode for remote development, e.g. **the integrated terminal** that allows you to run commands directly on the Raspberry Pi or the Git integration.

Note

It is now a good idea to install VSCode extensions that can help during development, e.g. C/C++ and Python extensions.

6.2.4. (Optional) Setting up SSH Key-based Authentication

With the configuration above, every time you connect to the Pi using VSCode, you will be prompted to enter the password for the **rpi-student** user. To avoid this, you can set up SSH key-based authentication. To do this, first we need to generate an SSH key pair (public and private) **on the host machine (Ubuntu VM)**. Open a terminal in the Ubuntu VM and run the following command:

```
ssh-keygen -t rsa -b 4096
```

Press Enter to accept the default values. After this a key pair will be generated in the *~/.ssh/* folder. Take a look at the generated files, and **copy the content of the public key** file (*id_rsa.pub*).

Now, we need to add the public key to the Raspberry Pi to allow key-based authentication. **Connect to the Raspberry Pi** using VSCode or a terminal and open (or create) the *~/.ssh/authorized_keys* file. Then paste the content of the public key file at the end of the *authorized_keys* file and save it.

Now you should be able to connect to the Raspberry Pi using VSCode without being prompted for a password.

Note

If you have issues connecting to the Raspberry Pi, or keeps asking for a password, try adding the key to the SSH agent (on the Ubuntu VM):

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_rsa
```

6.2.5. Compiling C/C++ Applications on the Raspberry Pi

Now that we have set up VSCode for remote development on the Raspberry Pi, we can compile and debug C/C++ applications directly on the Pi.

Important

Make sure that VSCode is connected to the Raspberry Pi as explained in the previous section.

First, lets create a new folder on the Raspberry Pi to create an example C application, we will be using the Makefile-based build system. Open the integrated terminal in VSCode (View -> Terminal) and run the following commands:

```
mkdir hello-world
```

Now, in VSCode, open the *hello-world* folder that we just created on the Raspberry Pi (File -> Open Folder...). The folder is currently empty, create a new file named *main.c* and add the following code:

```c
#include <stdio.h>

int main(void) {
    printf("Hello world from Raspberry Pi\n");
    return 0;
}
```

Next, create a new file named *Makefile* in the same folder and add the following content:

```
TARGET = main
SRCS = main.c
OBJS = $(SRCS:.c=.o)

INCLUDE =
LIBS = -lm -lpthread

CC ?= gcc
CFLAGS += -Wall -g $(INCLUDE)
LDFLAGS += $(LIBS)

all: $(TARGET)

$(TARGET): $(OBJS)
    @echo "[link] $^"
    $(CC) -o $@ $^ $(LDFLAGS)

%.o: %.c
    @echo "[compile] $<"
    $(CC) $(CFLAGS) -c $< -o $@

.PHONY: clean
clean:
    rm -f $(TARGET) $(OBJS)
```

Note

Check the flags used in the Makefile and note that the debug flag (-g) is enabled to allow debugging the application later.

Note

You can use this Makefile as a template for your future C applications.

Your file explorer should look like shown in figure Fig. 6.6.

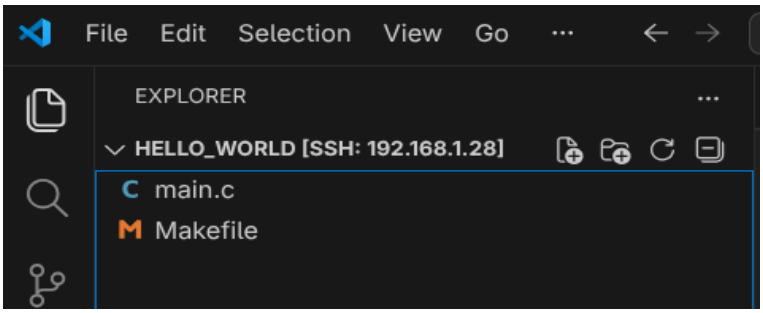


Fig. 6.6 Hello World project folder structure

Now, to compile the application, open the integrated terminal in VSCode and run the following command:

```
make
```

And run the application with:

```
./main
```

You should see something like shown in figure Fig. 6.7.

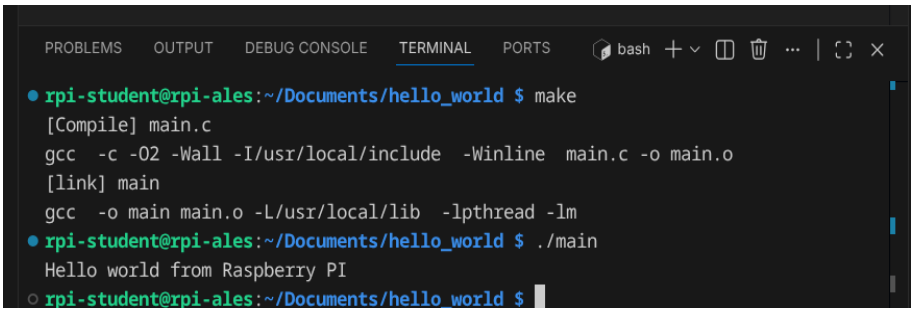


Fig. 6.7 Hello World application running on Raspberry Pi

In VSCode we can automate this process by creating tasks for building and running the application. In the file explorer, create a new folder named *.vscode* inside the *hello-world* folder. Then, inside the *.vscode* folder, create a new file named *tasks.json* and add the following content:

```
{
    "version": "2.0.0",
    "tasks": [
        {
            "label": "build",
            "type": "shell",
            "command": "make"
        },
        {
```

```
            "label": "build and run",
            "type": "shell",
            "command": "make && ./main"
        },
        {
            "label": "clean",
            "type": "shell",
            "command": "make clean"
        }
    ]
}
```

Your file explorer should look like shown in figure Fig. 6.8.

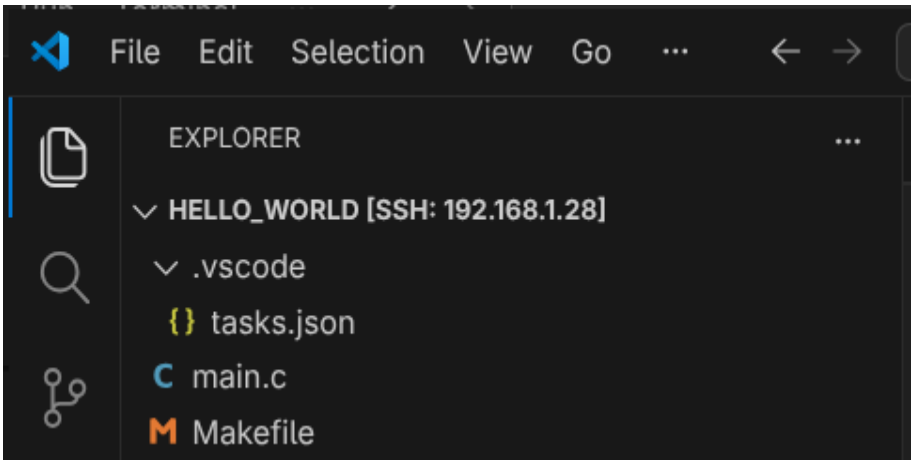


Fig. 6.8 VSCode tasks configuration

Now, you can use the "*build*" and "*build and run*" tasks to compile and run the application. To do this, in VSCode select *Terminal -> Run Task...* and select the desired task.

6.2.6. Debugging C/C++ Applications on the Raspberry Pi

VSCode also allows us to debug C/C++ applications running on the Raspberry Pi. To do this, we need to create a debug configuration. In the file explorer, inside the *.vscode* folder, create a new file named *launch.json* and add the following content:

```
{
    "version": "0.2.0",
    "configurations": [
        {
            "name": "Debug Raspberry Pi",
            "type": "cppdbg",
            "request": "launch",
            "program": "${workspaceFolder}/main",
            "args": [],
            "stopAtEntry": false,
```

```
            "cwd": "${workspaceFolder}",
            "environment": [],
            "externalConsole": false,
            "MIMode": "gdb",
            "setupCommands": [
                {
                    "description": "Enable pretty-printing for gdb",
                    "text": "-enable-pretty-printing",
                    "ignoreFailures": true
                }
            ],
            "preLaunchTask": "build",
            "miDebuggerPath": "/usr/bin/gdb"
        }
    ]
}
```

Note

Take your time to investigate the different options available in the debug configuration file so you can customize it to your projects.

Your file explorer should now look like shown in figure Fig. 6.9.

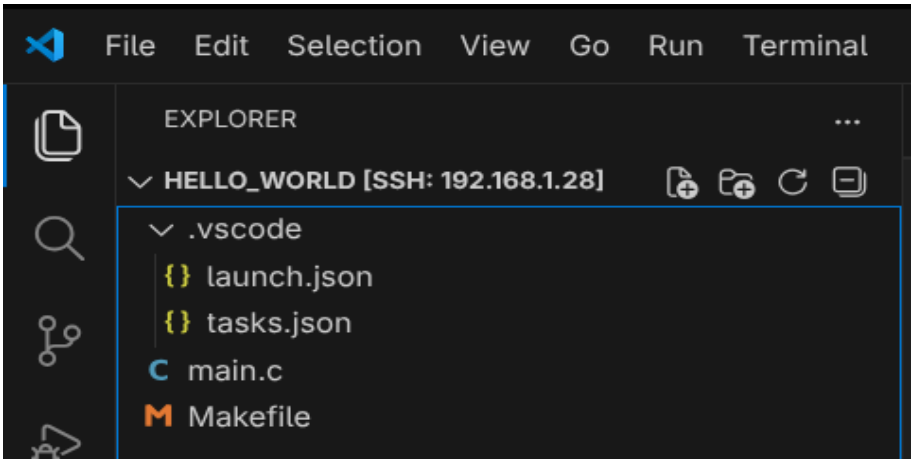


Fig. 6.9 VSCode debug configuration

Now, to start debugging the application, set a breakpoint in the *main.c* file by clicking on the left margin next to the line number. Then, in VSCode select *Run -> Start Debugging* (or press F5). The debug window will open and the application will stop at the breakpoint (see Fig. 6.10).

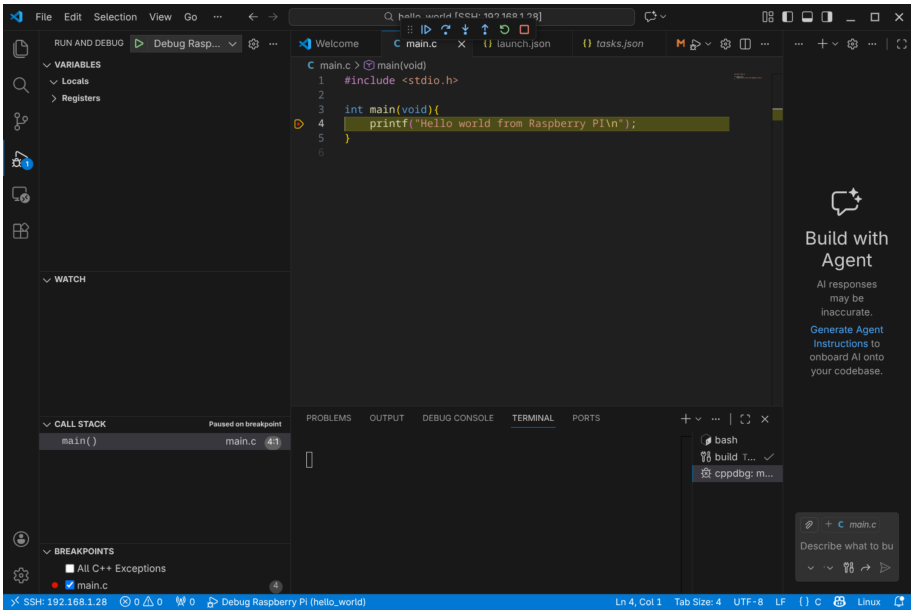


Fig. 6.10 VSCode debugging session

Notice the debug toolbar that allows you to control the debugging session (continue, step over, step into, etc.) and the different debug panels (variables, watch, call stack, breakpoints, etc.) available.

6.3. Setting up VSCode for Cross-Compiling Applications

We have seen how to set up VSCode for remote development, compiling and debugging applications natively on the Raspberry Pi. In this section we will see how to set up VSCode for cross-compiling applications for our custom embedded Linux system built with Buildroot.

Important

This section assumes that you have already generated the custom embedded Linux image as well as the cross-compile toolchain.

In this case we will be running VSCode on the host machine (Ubuntu VM) and we will use the buildroot-generated cross-compile toolchain to compile applications for the Raspberry Pi. We will copy the compilated applications to the Raspberry Pi and run them there.

Important

Make sure that you have installed the **C/C++ extension by Microsoft** in VSCode.

6.3.1. Configuring the VSCode project for Cross-Compilation

Start by creating a new folder for your project **on the host machine** and open it with VSCode (File -> Open Folder...).

Now, add the source files for your application to the project folder. For example, create a file named **main.c** with the basic *Hello World* shown in the previous section. Also, create a **Makefile** for your project. You can use the same Makefile shown in the previous section.

VSCode does not have built-in support for cross-compilation, but we can configure it to use the Buildroot-generated toolchain. To do so, create a folder named *.vscode* inside your project folder. Here is where we will add the configuration files for building and debugging our application.

Create inside the *.vscode* folder a file named *settings.json*. Add the following content to the file:

```
{
    "buildroot_build_dir": "<path_to_buildroot_build_directory>",
    "target_ip": "<rpi_ip_address>",
    "executable_file": "<executable_file_name>",
    "buildroot_env": "${config:buildroot_build_dir}/host/environment-
setup",
    "host_gdb": "${config:buildroot_build_dir}/host/bin/aarch64-
linux-gdb",
}
```

Replace the values of the different fields with the appropriate values for your setup:

- `<path_to_buildroot_build_directory>`: The path to the Buildroot build directory where the toolchain and sysroot are located, e.g. */home/ubuntu/Documents/rpi/build*.
- `<rpi_ip_address>`: The IP address of your Raspberry Pi.
- `<executable_file_name>`: The name of the executable file that you want to compile and run on the Raspberry Pi, e.g. *main*.

Now, lets configure the tasks that tell VSCode how to build and deploy our application. Create a file named *tasks.json* inside the *.vscode* folder and add the following content:

```
{
    "version": "2.0.0",
    "tasks": [
        {
```

```
            "label": "Build",
            "type": "shell",
            "command": "/bin/bash",
            "args": [
                "-c",
                "source ${config:buildroot_env} && make clean all"
            ],
            "group": {
                "kind": "build",
                "isDefault": true
            }
        },
        {
            "label": "Deploy",
            "type": "shell",
            "command": "ssh root@${config:target_ip} 'rm -f /root/$
{config:executable_file}' && scp ${config:executable_file} root@$
{config:target_ip}:/root/",
            "dependsOn": ["Build"]
        },
        {
            "label": "Start GDBServer",
            "type": "shell",
            "command": "ssh root@${config:target_ip} 'killall -q
gdbserver; gdbserver :2345 /root/${config:executable_file}'",
            "isBackground": true,
            "problemMatcher": {
                "pattern": { "regexp": "." },
                "background": {
                    "activeOnStart": true,
                    "beginsPattern": "Process /root/.* created",
                    "endsPattern": "Listening on port 2345"
                }
            },
            "dependsOn": ["Deploy"]
        },
        {
            "label": "Kill GDBServer on Target",
            "type": "shell",
            "command": "ssh root@${config:target_ip} 'killall -q
gdbserver || true'",
            "problemMatcher": []
        }
    ]
}
```

This configuration defines the following tasks:

- `Build`: This task builds the application using the Buildroot-generated toolchain. First, it **sources the Buildroot environment setup script to set up the necessary environment variables for cross-compilation**, and then it uses the Makefile to build the application.

- `Deploy`: This task deploys the compiled application to the Raspberry Pi. It does so by first removing any existing executable with the same name on the Raspberry Pi and then copying the new executable using SCP (using the SSH protocol).
- `Start GDBServer`: This task starts a GDB server on the Raspberry Pi to allow remote debugging.
- `Kill GDBServer on Target`: This task kills any running GDB server on the Raspberry Pi.

Now, we need to configure the debug configuration to allow debugging the application running on the Raspberry Pi. Create a file named *launch.json* inside the *.vscode* folder and add the following content:

```
{
    "version": "0.2.0",
    "configurations": [
        {
            "name": "Debug on Target",
            "type": "cppdbg",
            "request": "launch",
            "program": "${config:executable_file}",
            "args": [],
            "stopAtEntry": true,
            "cwd": "${workspaceFolder}",
            "environment": [],
            "externalConsole": false,
            "MIMode": "gdb",
            "miDebuggerPath": "${config:host_gdb}",
            "miDebuggerServerAddress": "${config:target_ip}:2345",
            "preLaunchTask": "Start GDBServer",
            "postDebugTask": "Kill GDBServer on Target",
            "setupCommands": [
                {
                    "description": "Enable pretty-printing for gdb",
                    "text": "-enable-pretty-printing",
                    "ignoreFailures": true
                }
            ]
        }
    ]
}
```

This defines a debug configuration named **Debug on Target** that uses the GDB server running on the Raspberry Pi to debug the application. Some important fields in this configuration are:

- `program`: The name of the executable file to debug.
- `miDebuggerPath`: The path to the GDB executable on the host machine (the Buildroot-generated GDB). Remember that we are debugging from an x86_64 machine an application running on an ARM machine, so we need to use the Buildroot-generated GDB that supports cross-debugging.

- `miDebuggerServerAddress`: The address of the GDB server running on the Raspberry Pi (IP address and port).
- `preLaunchTask`: The task to run before starting the debug session (in this case, we start the GDB server, which also builds and deploys the application).

Note

You can customize this debug configuration to fit your needs, e.g. by adding arguments to the application (use the "args" field), setting environment variables ("environment" field), etc.

This is enough configuration to allow us to build, deploy and debug our application. However, you may have noticed that VSCode reports errors in the source code because it cannot find the header files and libraries for our target system. To fix this, and get proper code completion and error checking, we need to configure the C/C++ extension to use the Buildroot sysroot. To do this, create a file named *c_cpp_properties.json* inside the *.vscode* folder and add the following content:

```
{
    "configurations": [
        {
            "name": "Buildroot-RPI",
            "includePath": [
                "${workspaceFolder}/**",
                "${config:buildroot_build_dir}/host/aarch64-
buildroot-linux-gnu/sysroot/usr/include"
            ],
            "defines": [],
            "compilerPath": "${config:buildroot_build_dir}/host/bin/
aarch64-linux-gcc",
            "cStandard": "c11",
            "cppStandard": "c++17",
            "intelliSenseMode": "linux-gcc-arm",
            "compilerArgs": [
                "--sysroot=${config:buildroot_build_dir}/host/arm-
buildroot-linux-gnueabihf/sysroot"
            ],
            "browse": {
                "path": [
                    "${workspaceFolder}",
                    "${config:buildroot_build_dir}/host/aarch64-
buildroot-linux-gnu/sysroot/usr/include"
                ],
                "limitSymbolsToIncludedHeaders": true
            }
        }
    ],
```

```
    "version": 4
}
```

This tells the C/C++ extension to use the Buildroot sysroot for code completion and error checking. The most important fields are:

- `includePath`: This field specifies the include paths for header files. We add the Buildroot sysroot include directory to the include path.
- `compilerPath`: This field specifies the path to the compiler to use for code analysis. We set it to the Buildroot-generated cross-compiler.
- `compilerArgs`: This field allows us to specify additional arguments to pass to the compiler for code analysis. We use it to specify the sysroot to use for code analysis.

At this point your VSCode project should look like shown in figure Fig. 6.11.

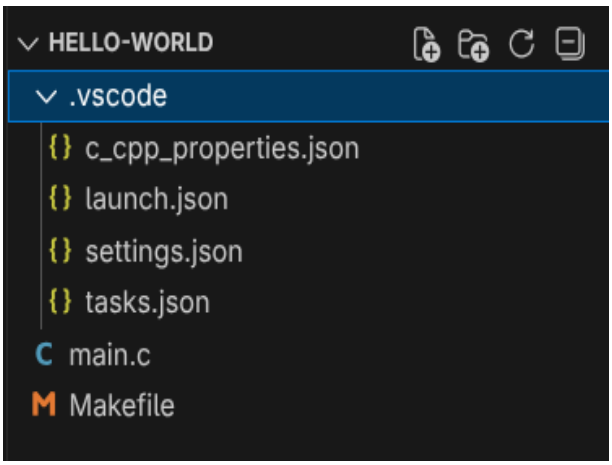


Fig. 6.11 VSCode project structure for cross-compilation

This is enough to allow you to build, deploy and debug your application for the Raspberry Pi using VSCode.

6.3.2. Setting up an SSH Key

This compilation, deployment and debugging process uses SSH to connect to the Raspberry Pi. You might be asked several times to enter the password for the **root** user on the Raspberry Pi. It is now **highly recommended** to set up SSH key-based authentication to avoid this. The process is almost the same as explained in the previous section (Optional) Setting up SSH Key-based Authentication :

1. Generate an SSH key pair on the host machine (Ubuntu VM) if you do not have one yet.

2. Copy the content of the public key file (e.g. *id_rsa.pub*).

3. Connect to the Raspberry Pi and add the public key to the */root/.ssh/authorized_keys* file (create it if it does not exist).

You should now be able to connect to the Raspberry Pi without being constantly prompted for a password.

6.3.3. Cross-Compiling, Deploying and Executing the Application

Now, use the defined tasks to build and deploy your application to the Raspberry Pi. To do this, in VSCode select *Terminal -> Run Task...*. On the top you should see a list of the defined tasks as shown in figure Fig. 6.12.

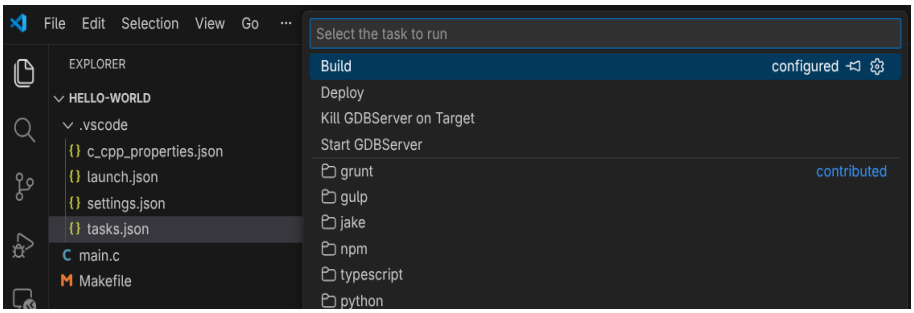


Fig. 6.12 List of defined tasks in VSCode

Select the *Deploy* task to build and deploy the application to the Raspberry Pi. You should see in the terminal output how the buildroot environment is sourced, the aplication is built, and then copied to the Raspberry Pi using SCP (see Fig. 6.13).

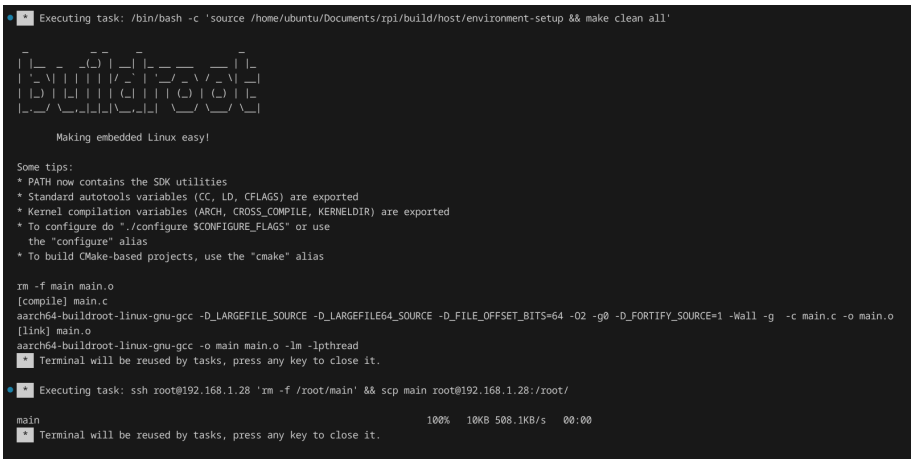


Fig. 6.13 Build and deploy task output in VSCode terminal

After the deployment is completed, you can connect to the Raspberry Pi using SSH and run the application executable, which can be found in the */root/* directory.

Note

The recommended flow is to use the *Build* task to build the application until you have no compilation errors, and then use the *Deploy* task to deploy the application to the Raspberry Pi.

Note

You can create additional tasks to automate other processes, e.g. a task to run the application on the Raspberry Pi using SSH.

6.3.4. Debugging the Application

If you want to debug the application, set a breakpoint in the source code (see Fig. 6.14) and then start the debug session (Run -> Start Debugging or F5). You will see in the terminal how the application is compiled, deployed and the GDB server is started on the Raspberry Pi. The debug view will open and you will be able to debug the application running on the Raspberry Pi (see Fig. 6.15). Then, you can use the debug toolbar to control the program execution, inspect variables...

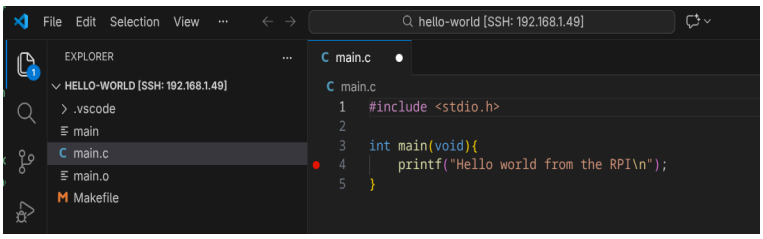


Fig. 6.14 Setting a breakpoint in VSCode

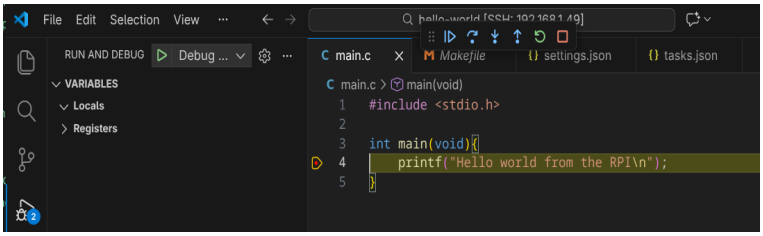


Fig. 6.15 Debugging an application running on the Raspberry Pi from VSCode

Indices and tables

Created using Sphinx 8.2.3.